



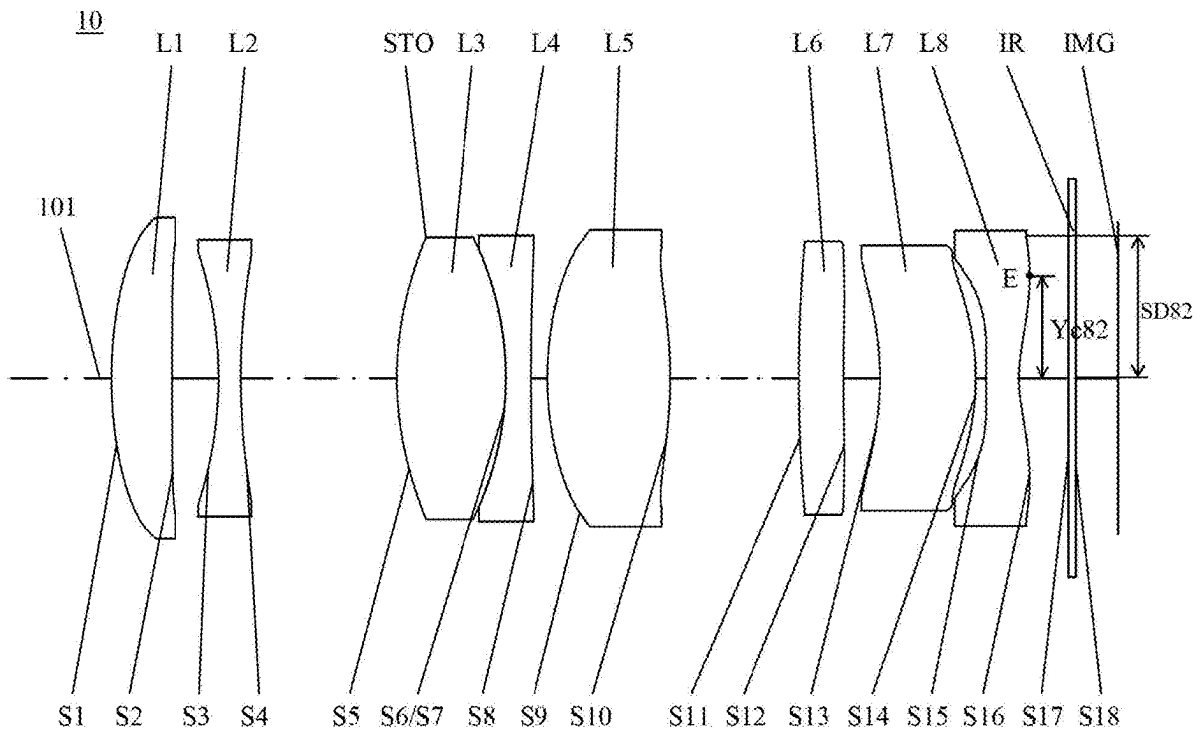
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(19) **United States**(12) **Patent Application Publication**
LIU(10) **Pub. No.: US 2025/0284097 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **OPTICAL SYSTEM, CAMERA MODULE,
AND ELECTRONIC DEVICE**(52) **U.S. Cl.**
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Nanchang (CN)(72) Inventor: **Binbin LIU**, Nanchang (CN)(21) Appl. No.: **19/070,826**(22) Filed: **Mar. 5, 2025**(30) **Foreign Application Priority Data**

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Publication Classification(51) **Int. Cl.**
G02B 13/00 (2006.01)
G02B 15/14 (2006.01)(57) **ABSTRACT**

An optical system consists of eight lenses with refractive power. From an object side to an image side along an optical axis of the optical system, the eight lenses sequentially include a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens. The first lens and the second lens form a first lens group with negative refractive power. The third lens to the fifth lens form a second lens group with positive refractive power. The sixth lens to the eighth lens form a third lens group having negative refractive power. The first lens group is fixed relative to an imaging plane of the optical system. Each of the second lens group and the third lens group can move along the optical axis between the first lens group and the imaging plane to realize zooming.



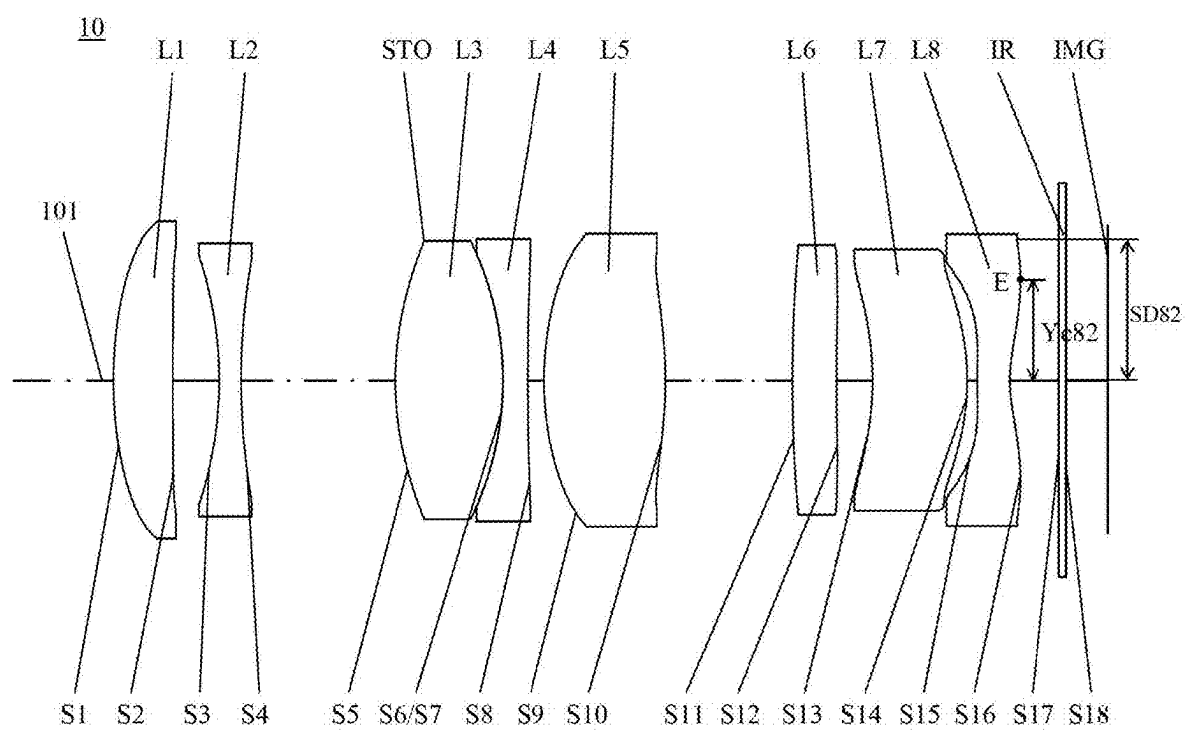
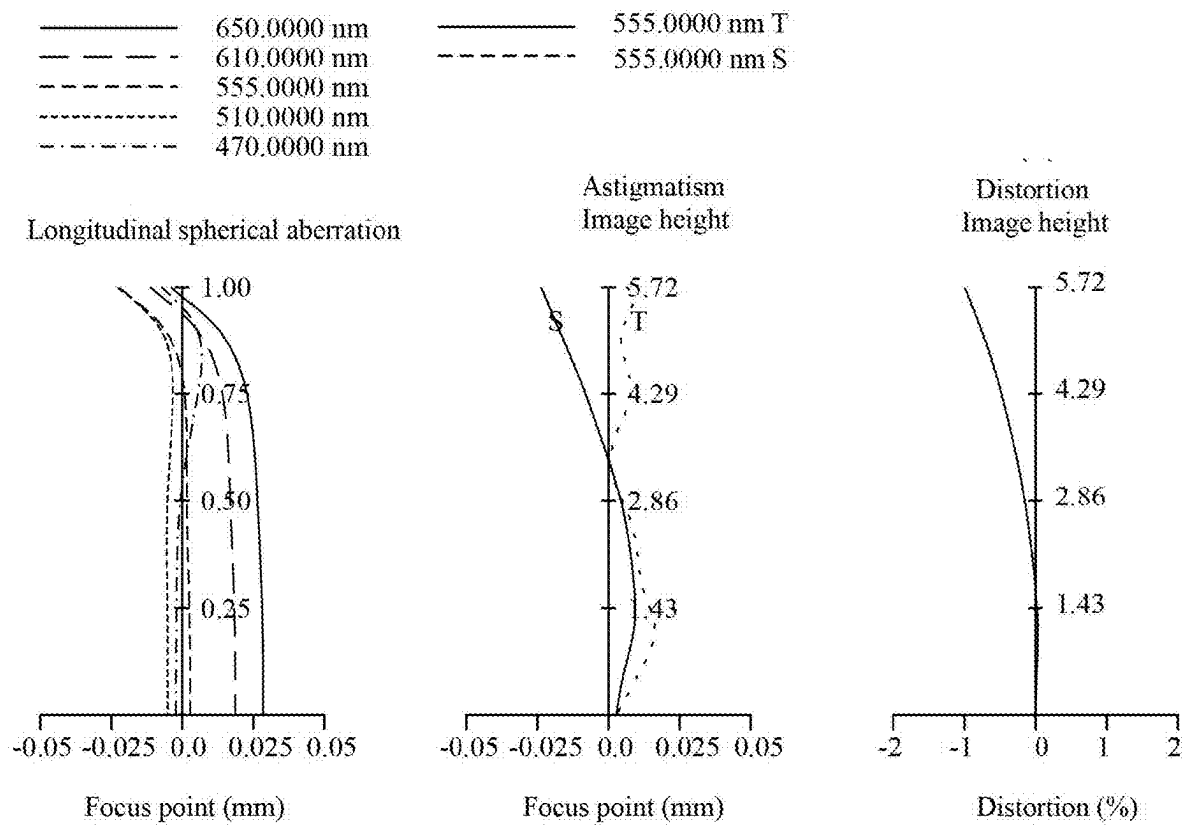


FIG. 1A



Distortion
Image height

5.72

4.29

2.86

1.43

0.00

-1.0

-2.0

-2

-1

0

1

2

Distortion (%)

FIG. 1B

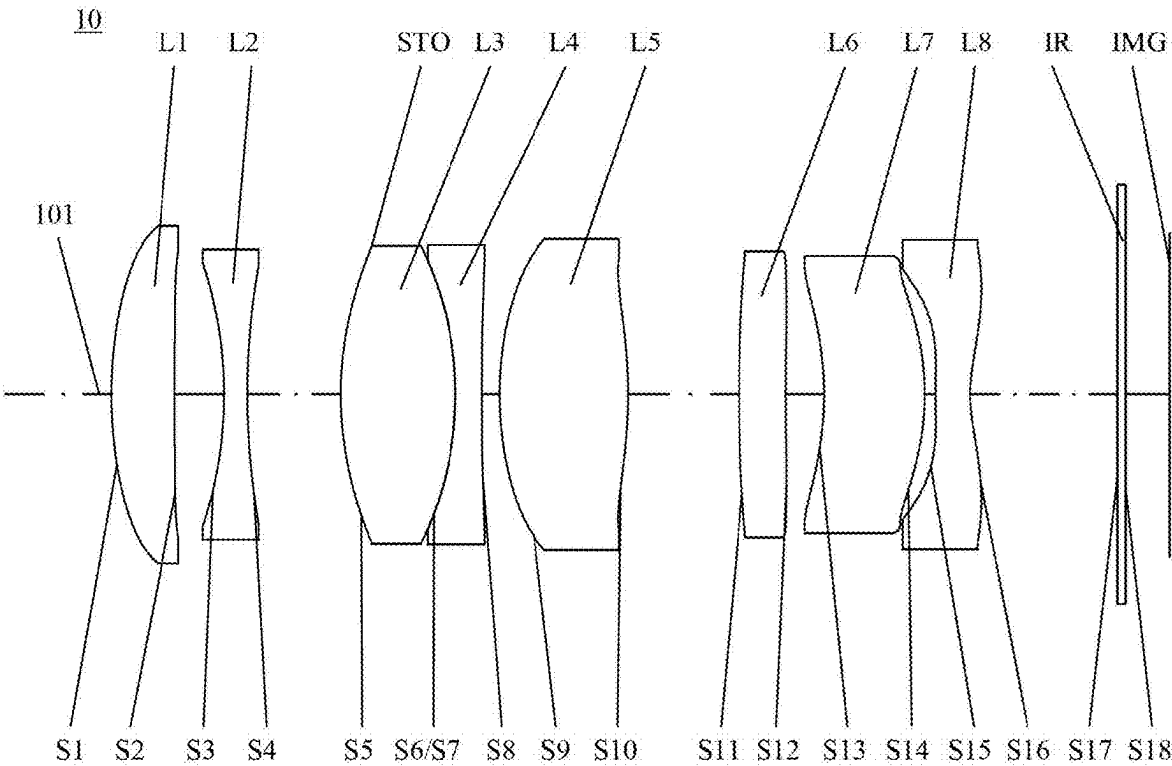


FIG. 1C

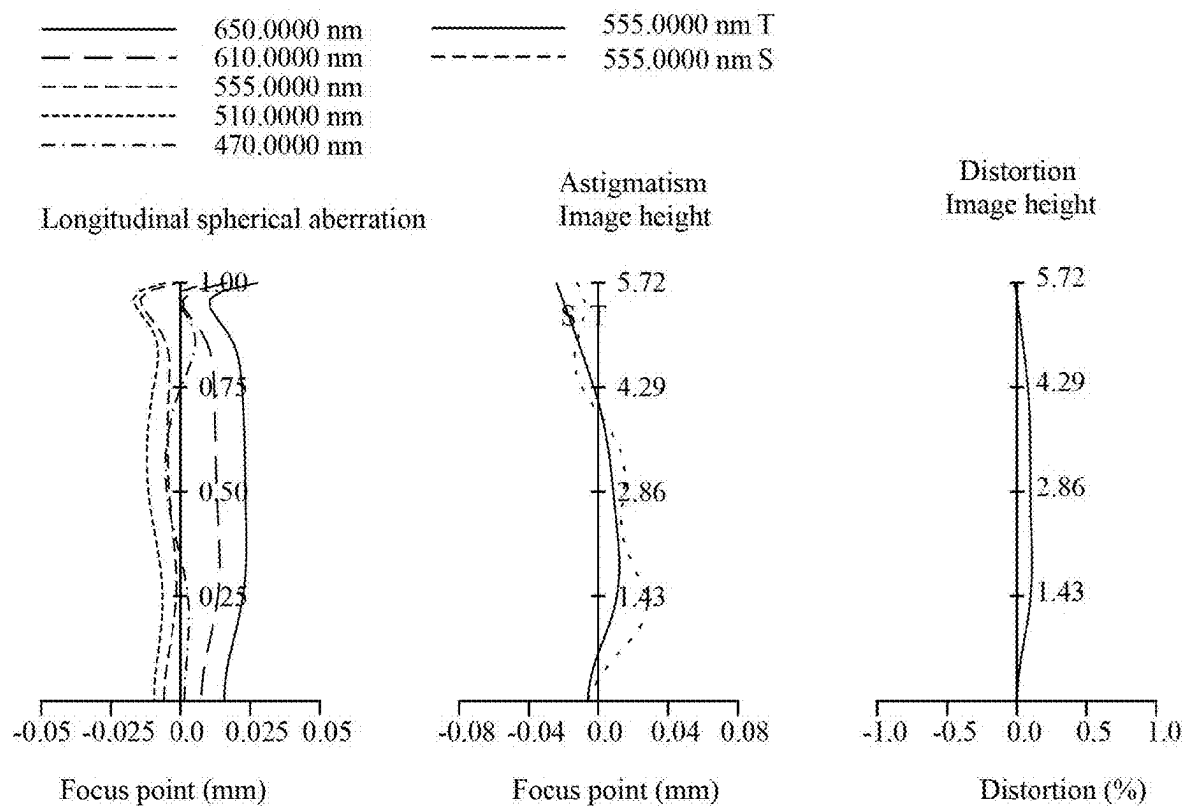


FIG. 1D

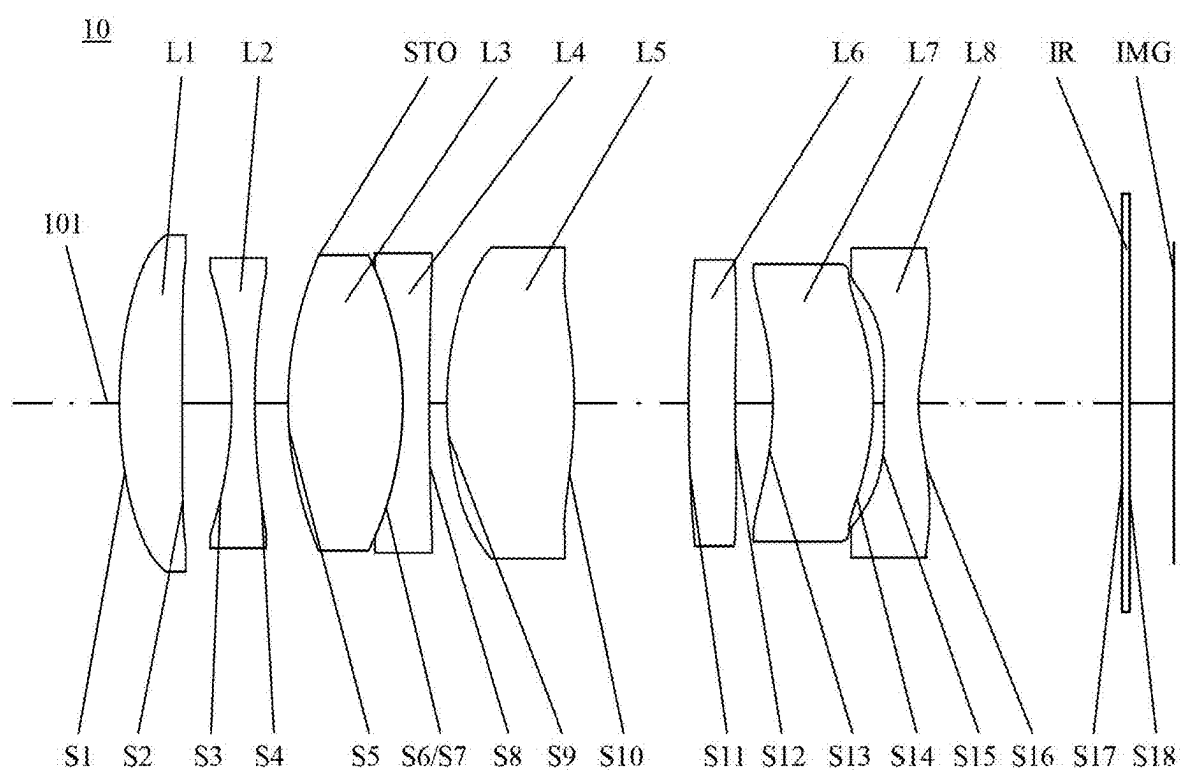


FIG. 1E

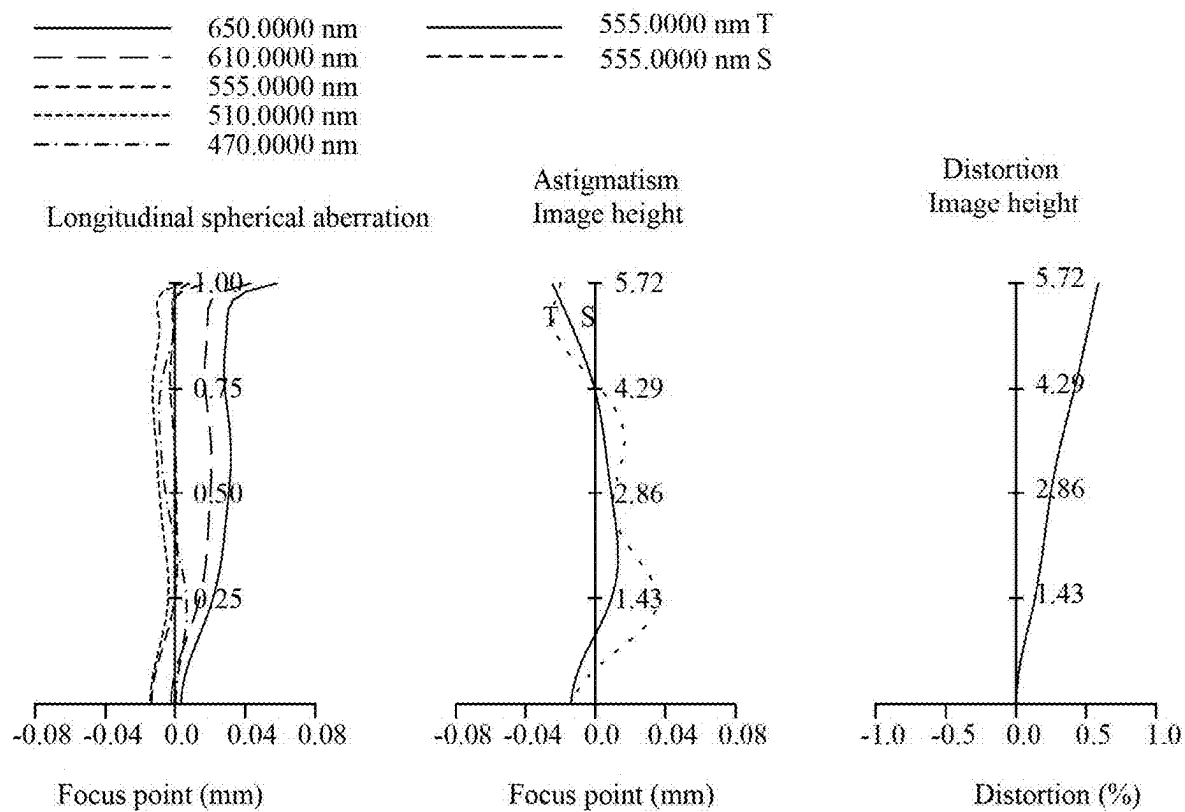


FIG. 1F

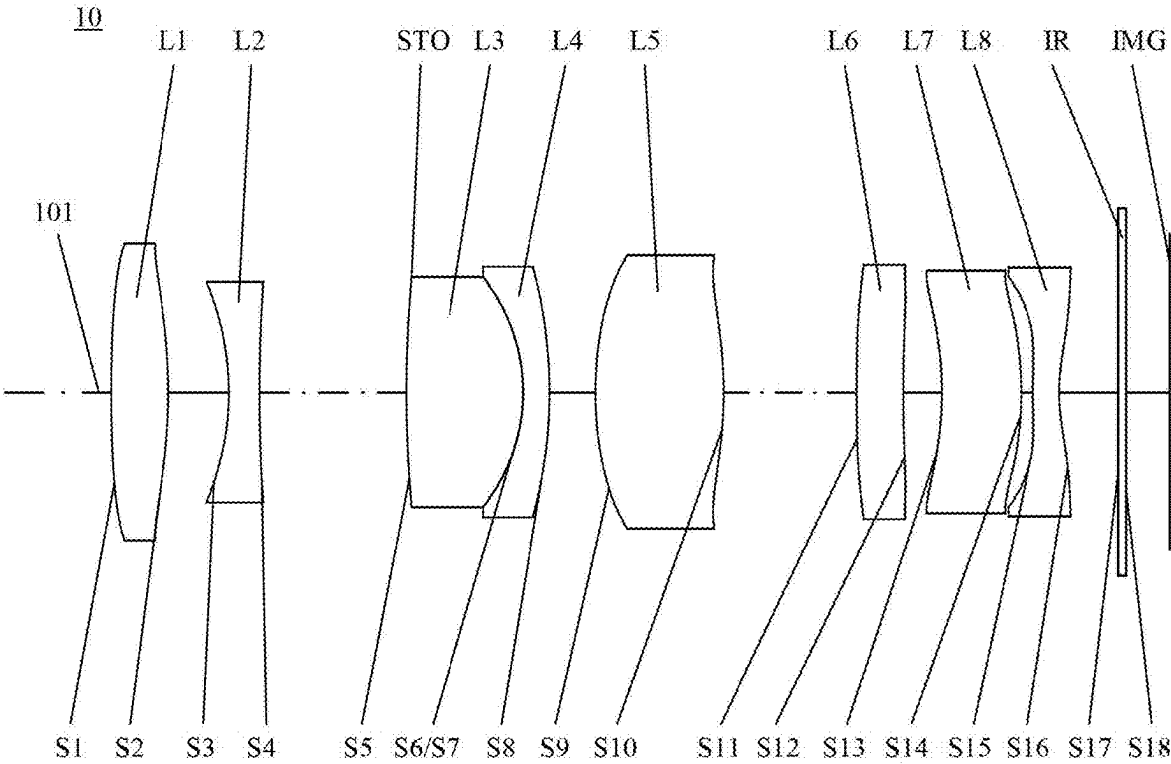
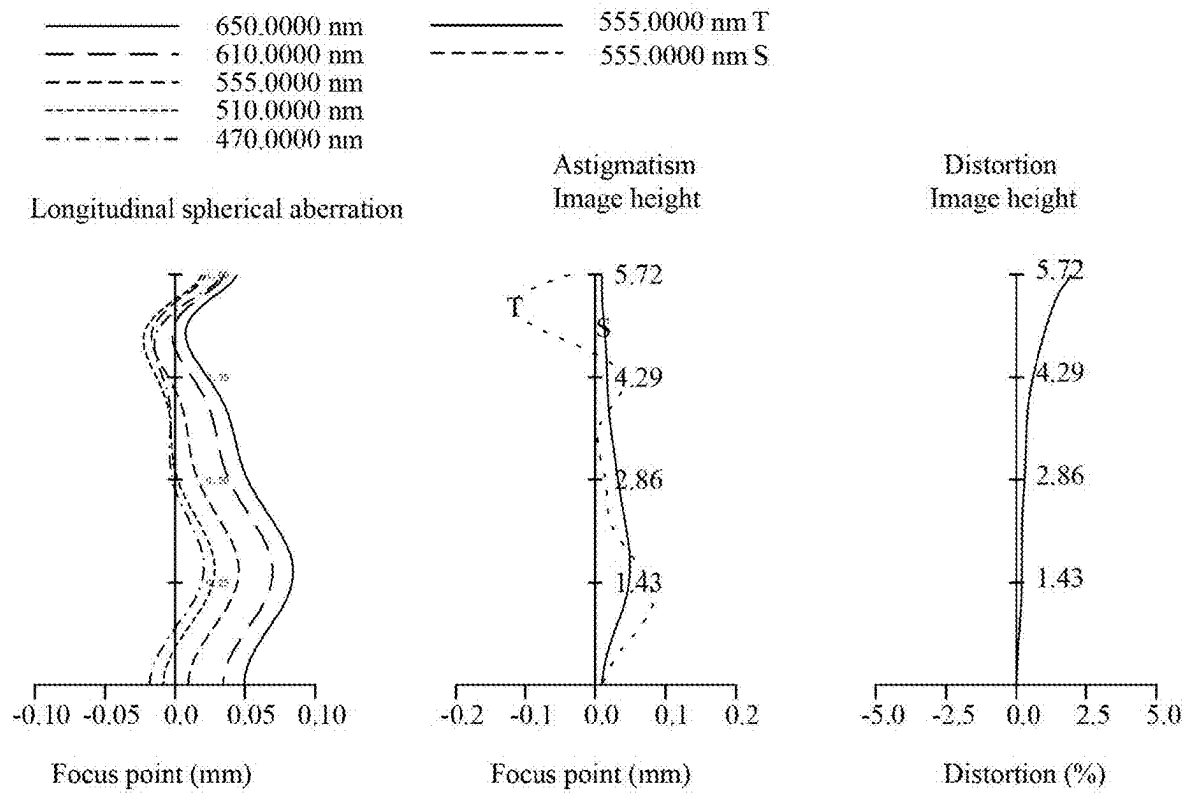


FIG. 2A



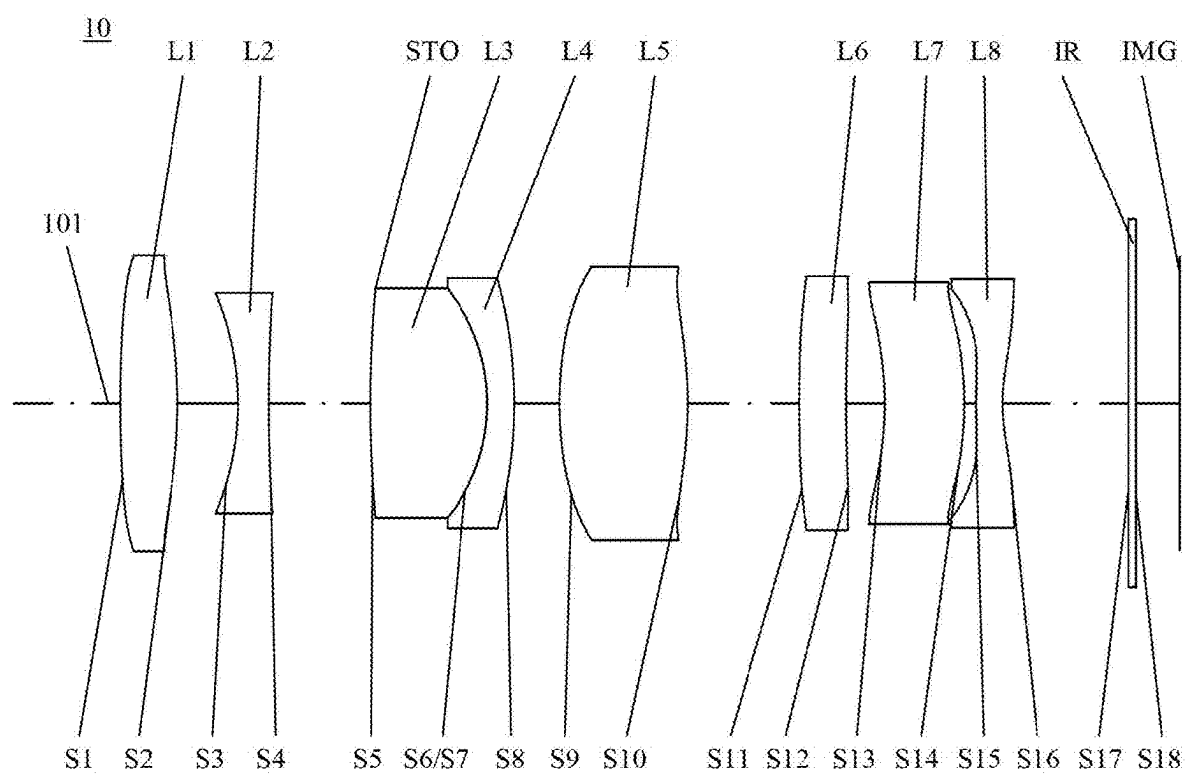
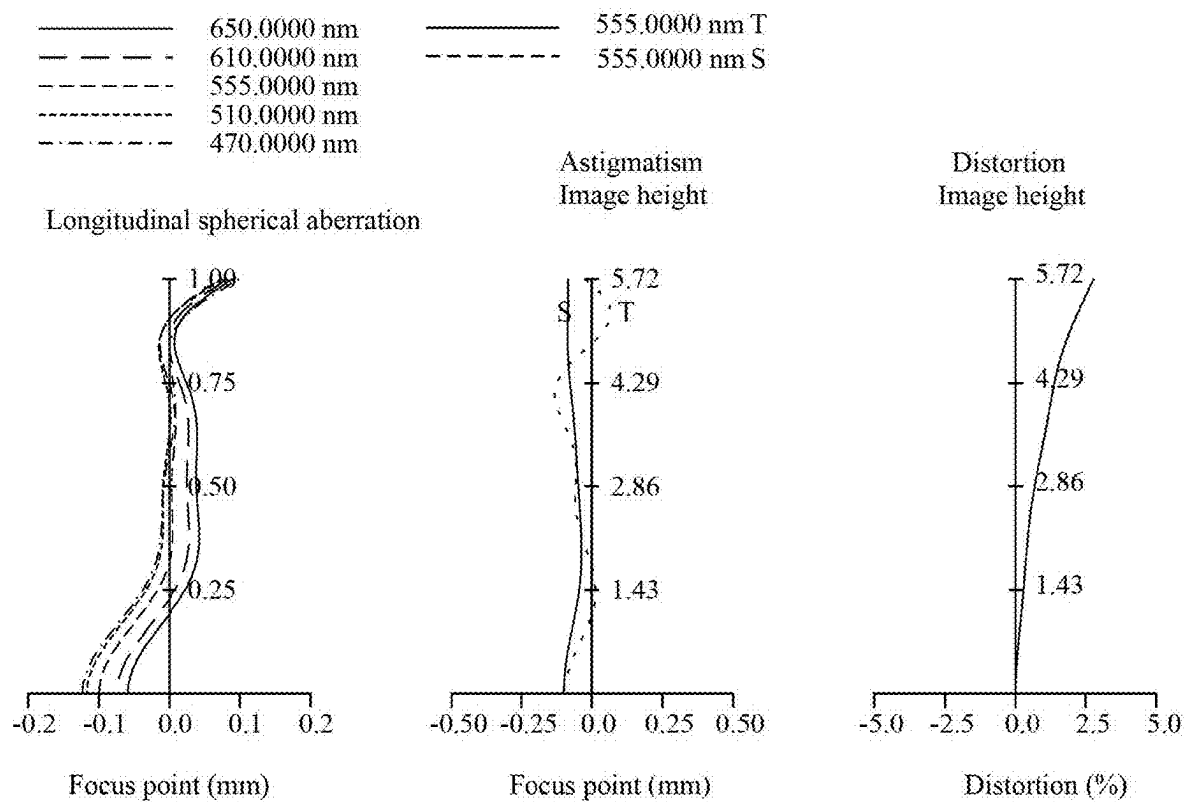


FIG. 2C



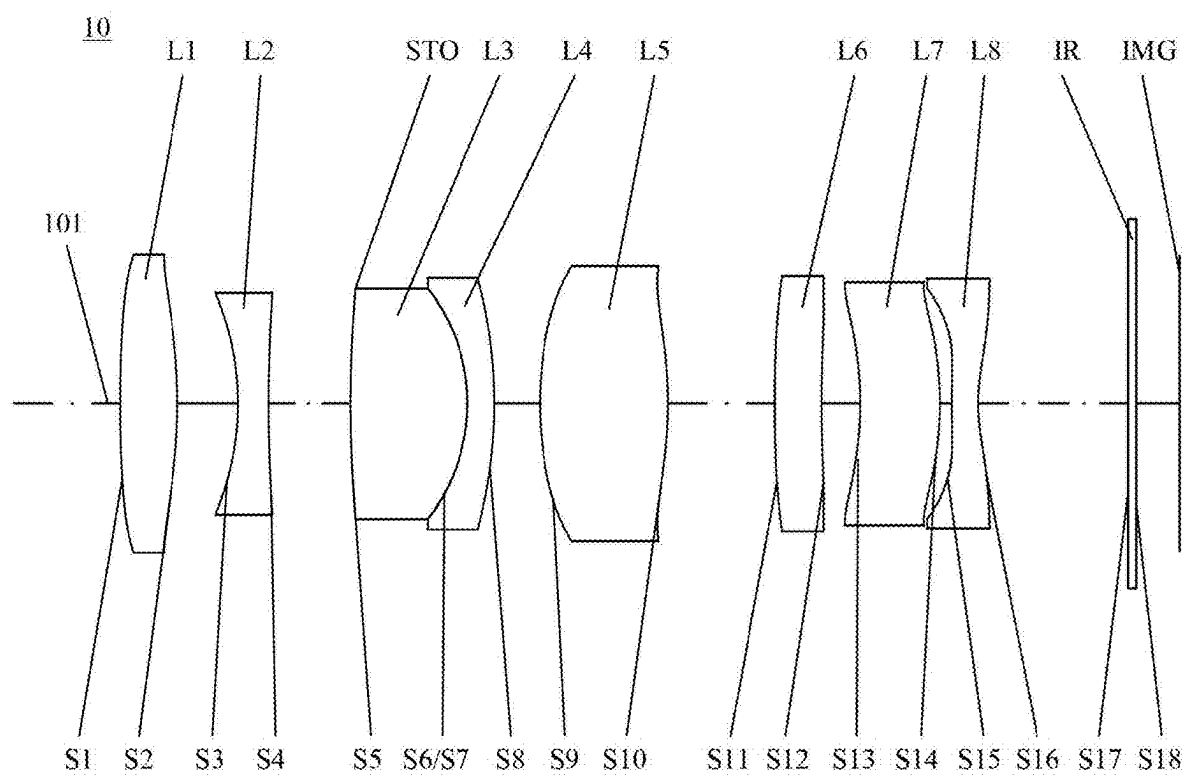


FIG. 2E

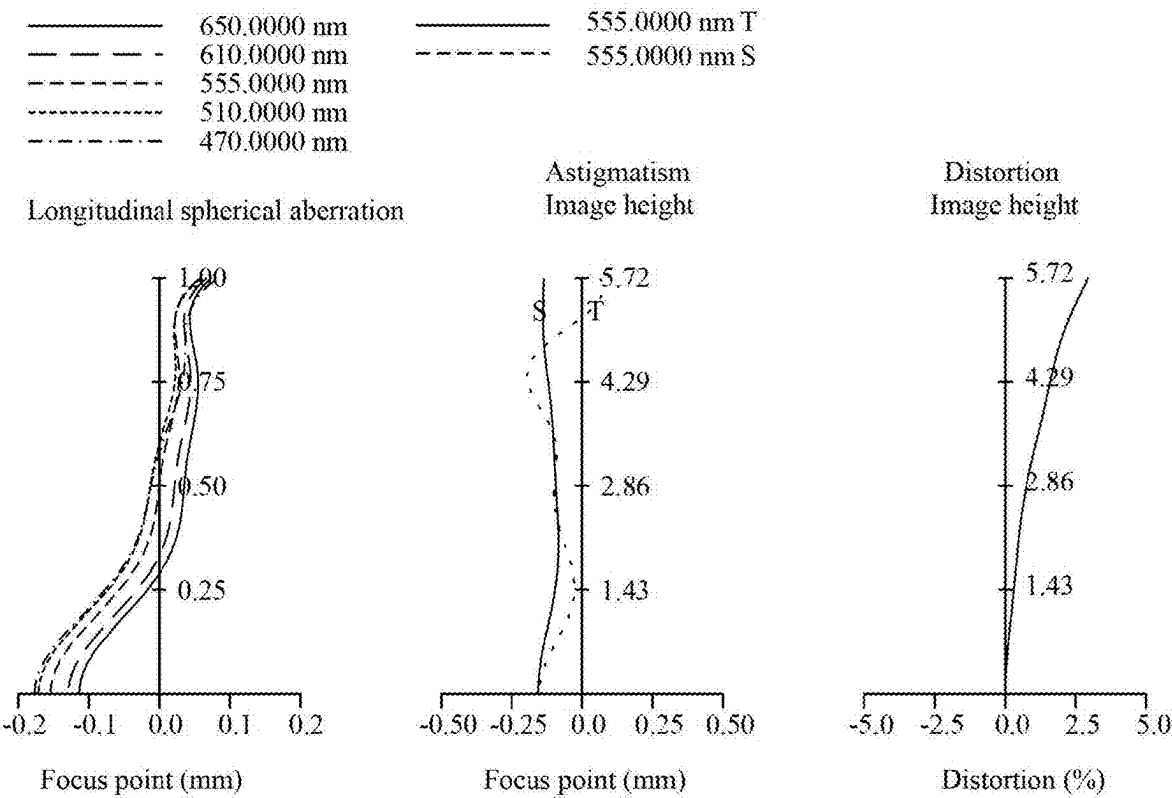


FIG. 2F

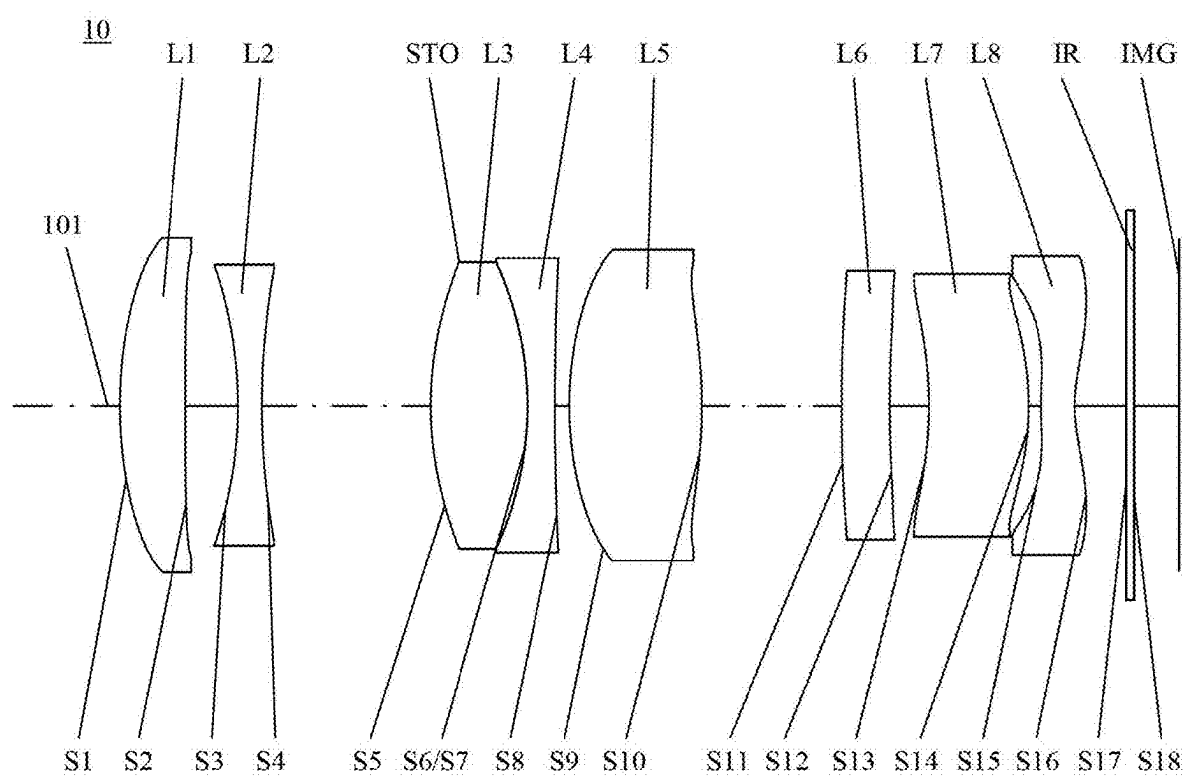


FIG. 3A

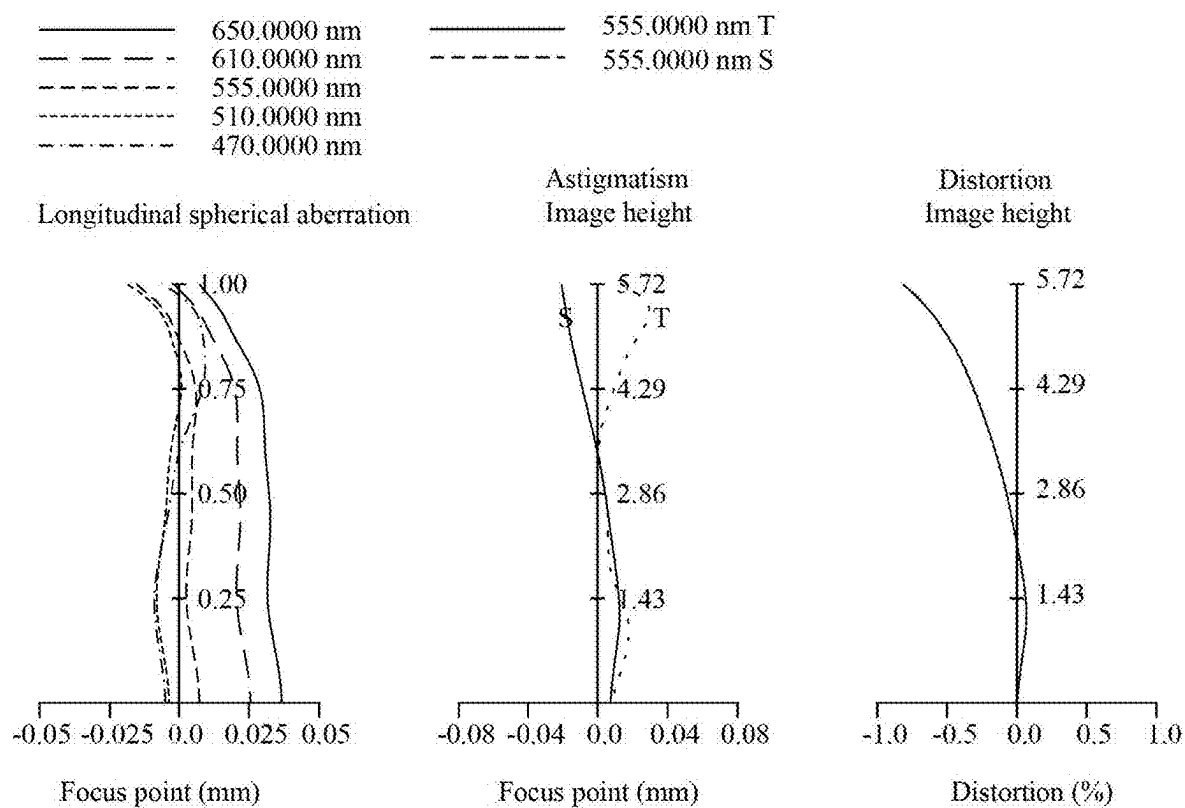


FIG. 3B

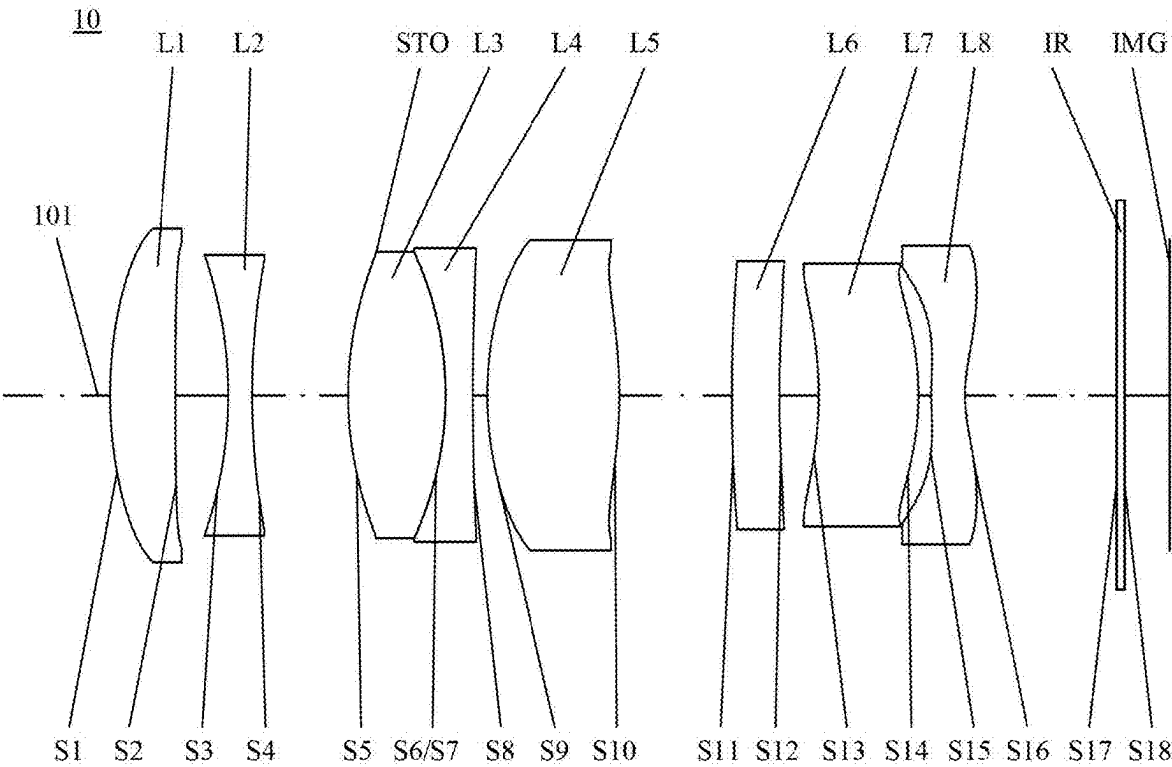
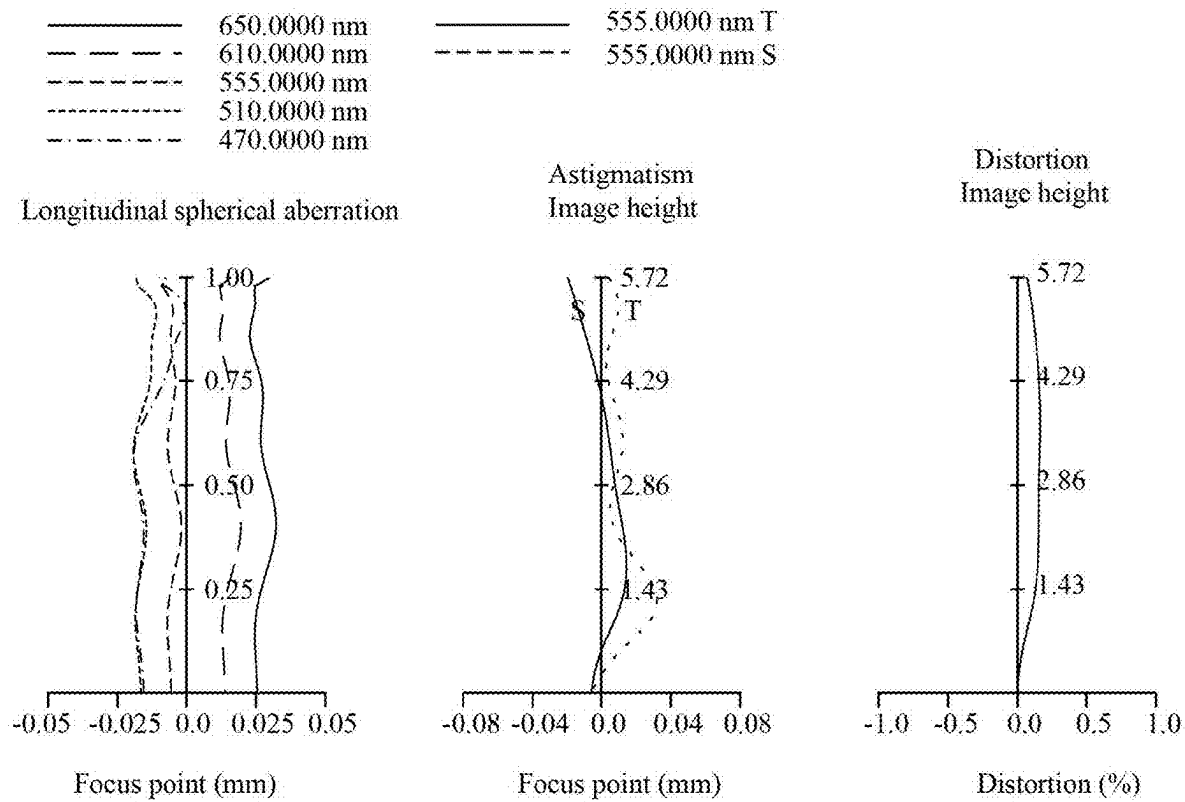


FIG. 3C



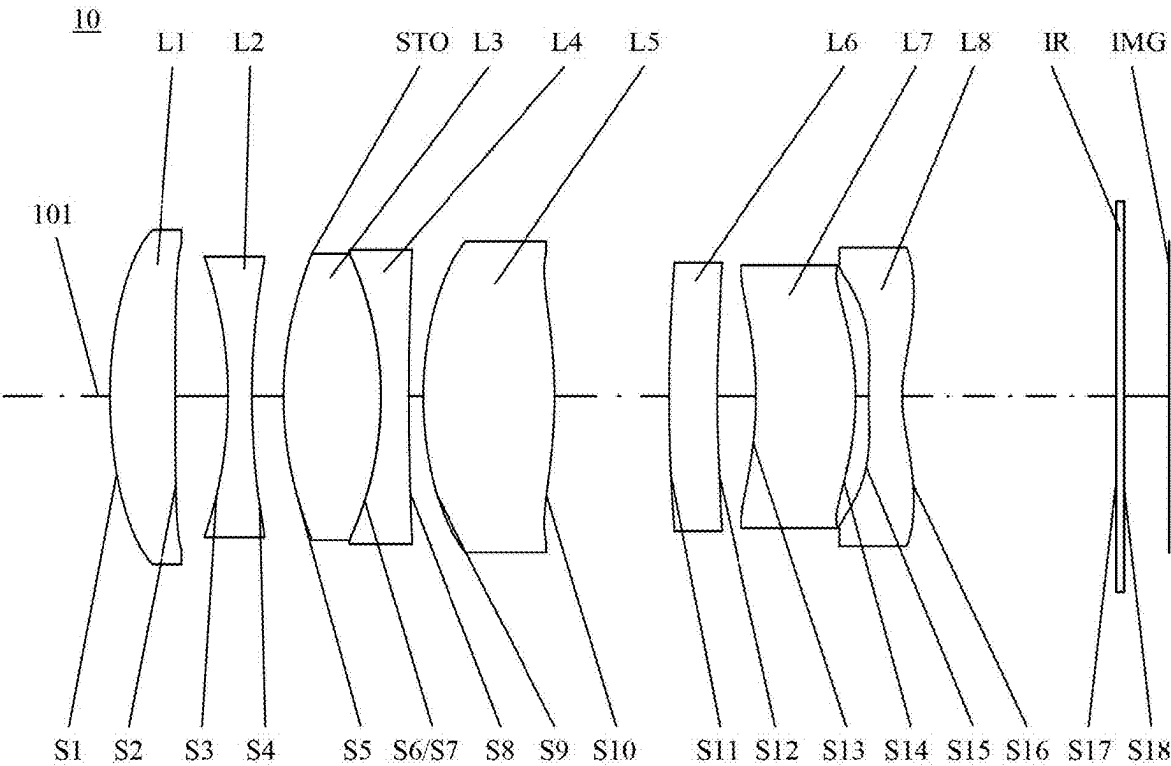
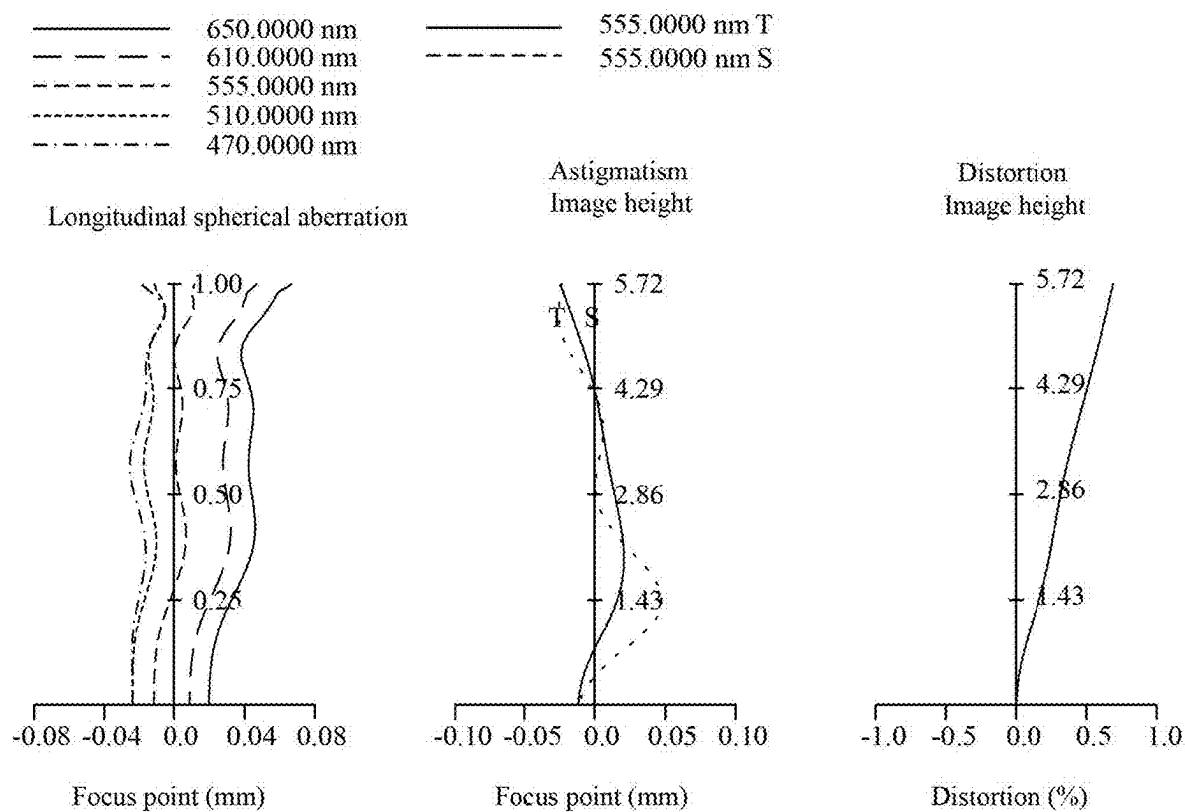


FIG. 3E



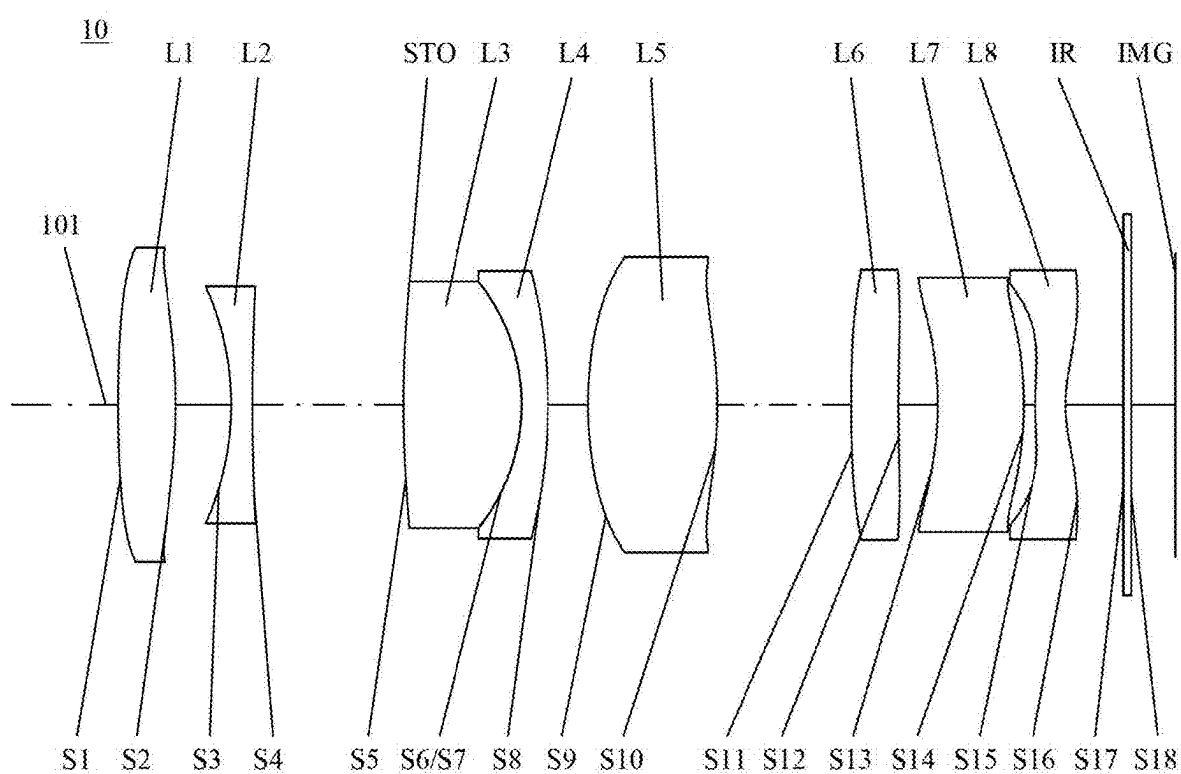
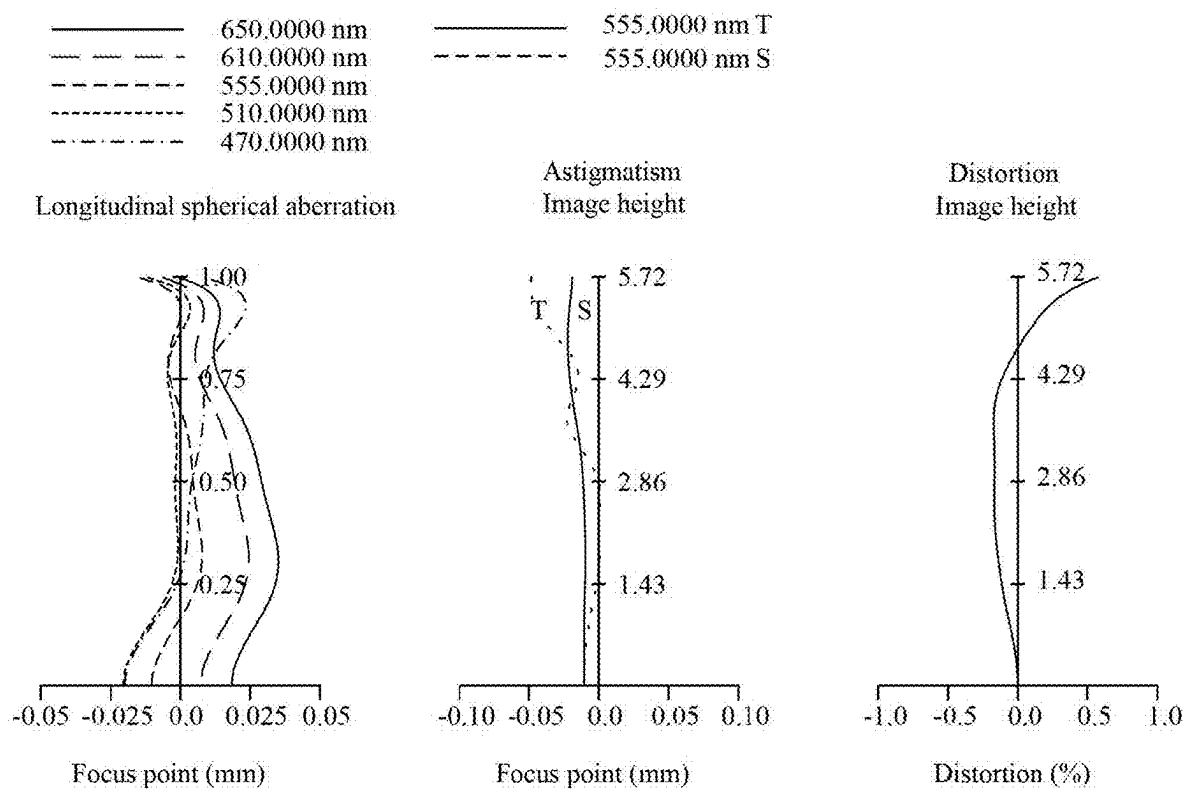


FIG. 4A



Astigmatism
Image height

Distortion
Image height

Focus point (mm)

Focus point (mm)

Distortion (%)

FIG. 4B

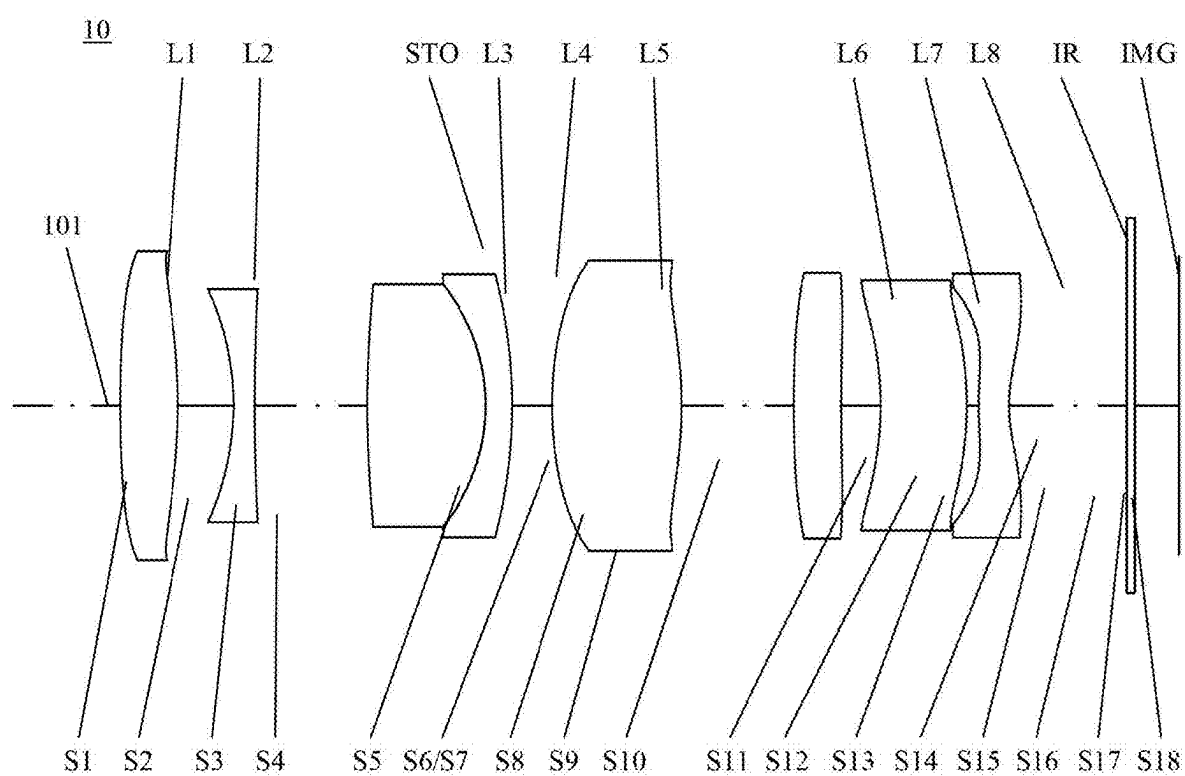


FIG. 4C

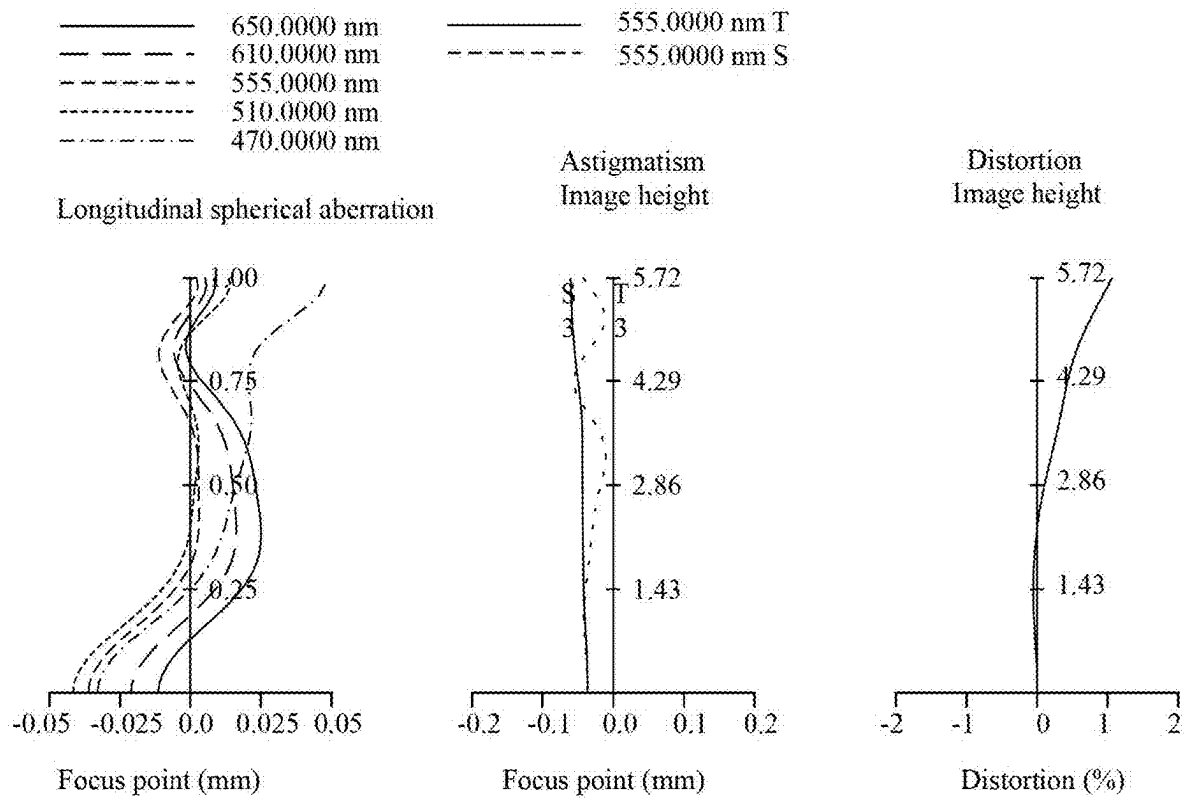


FIG. 4D

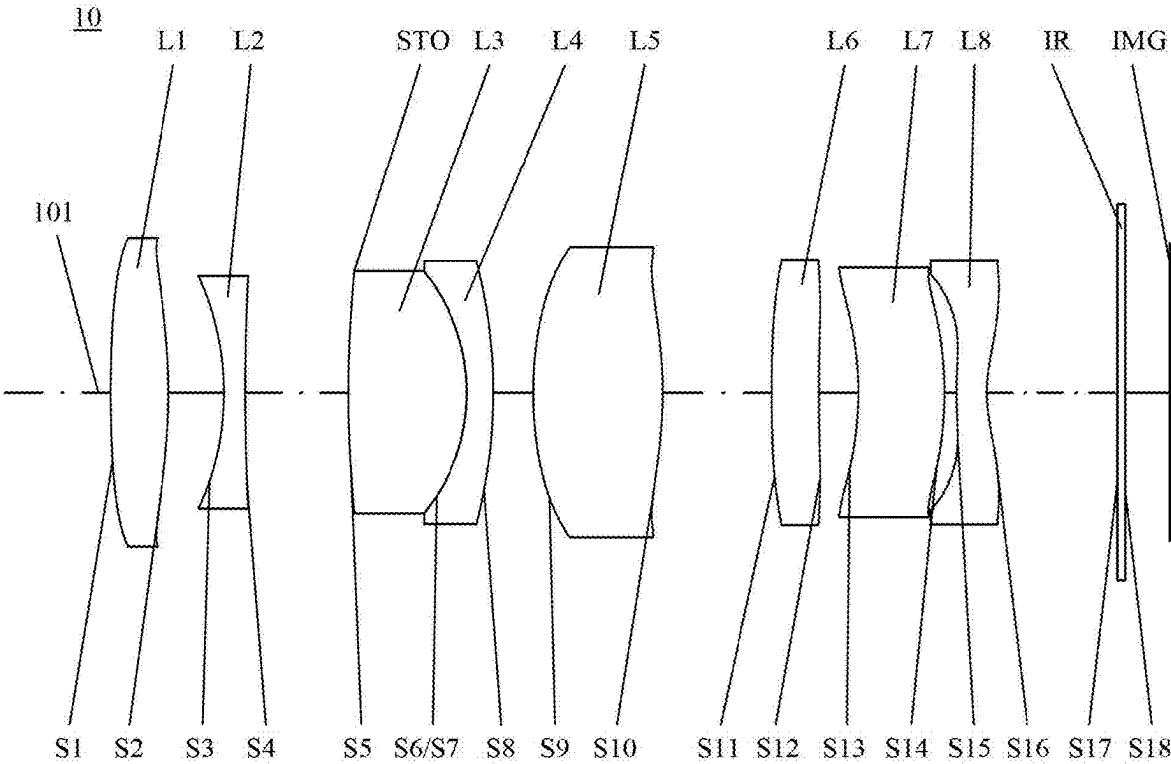
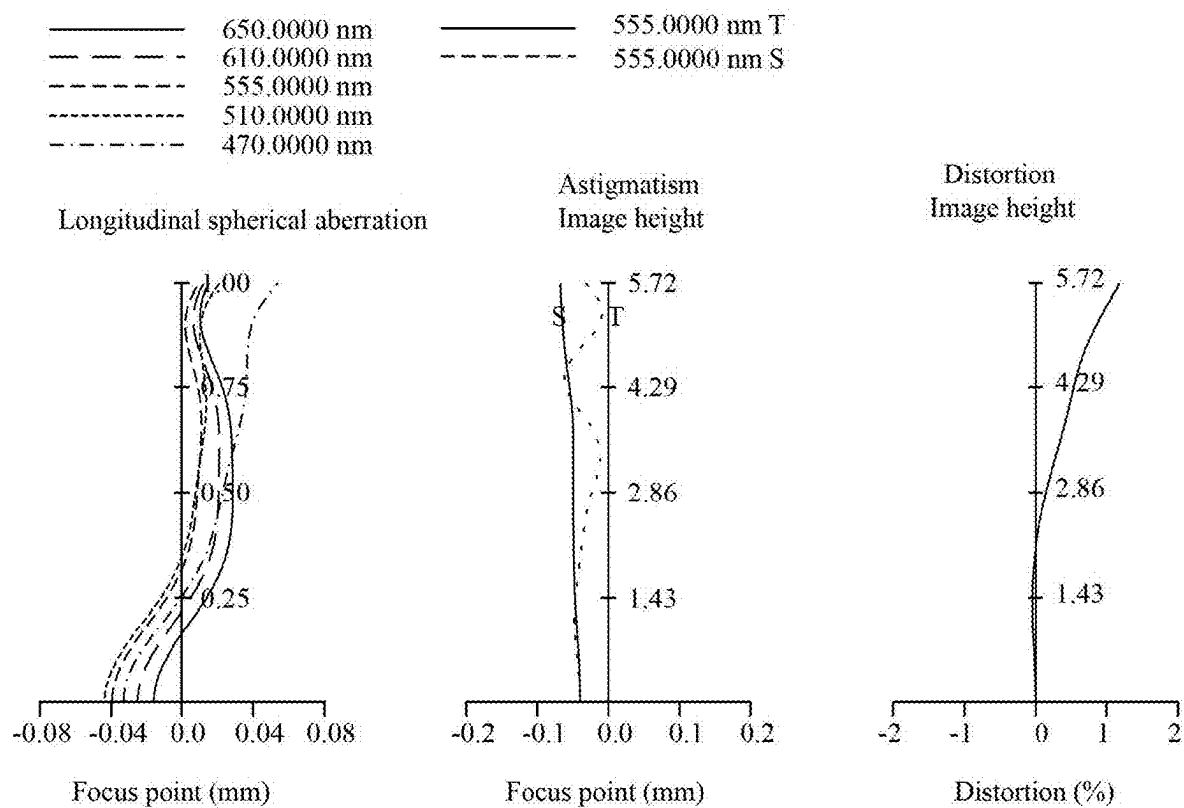


FIG. 4E



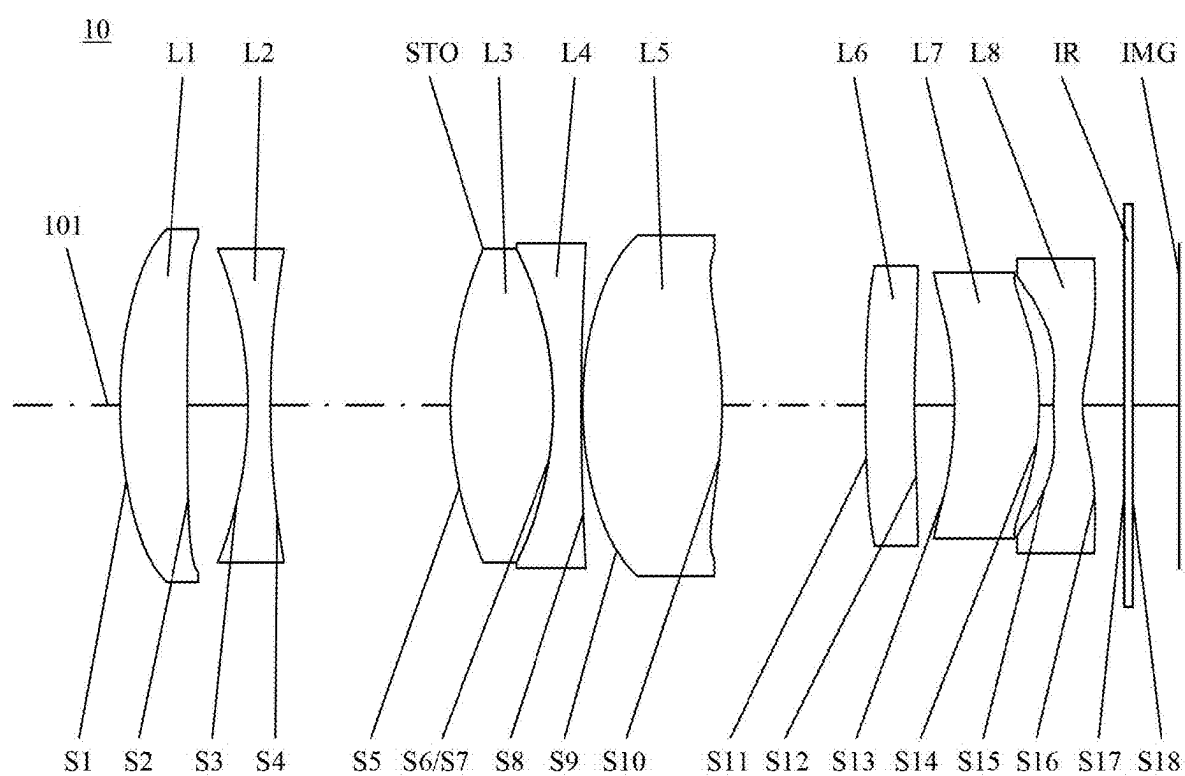


FIG. 5A

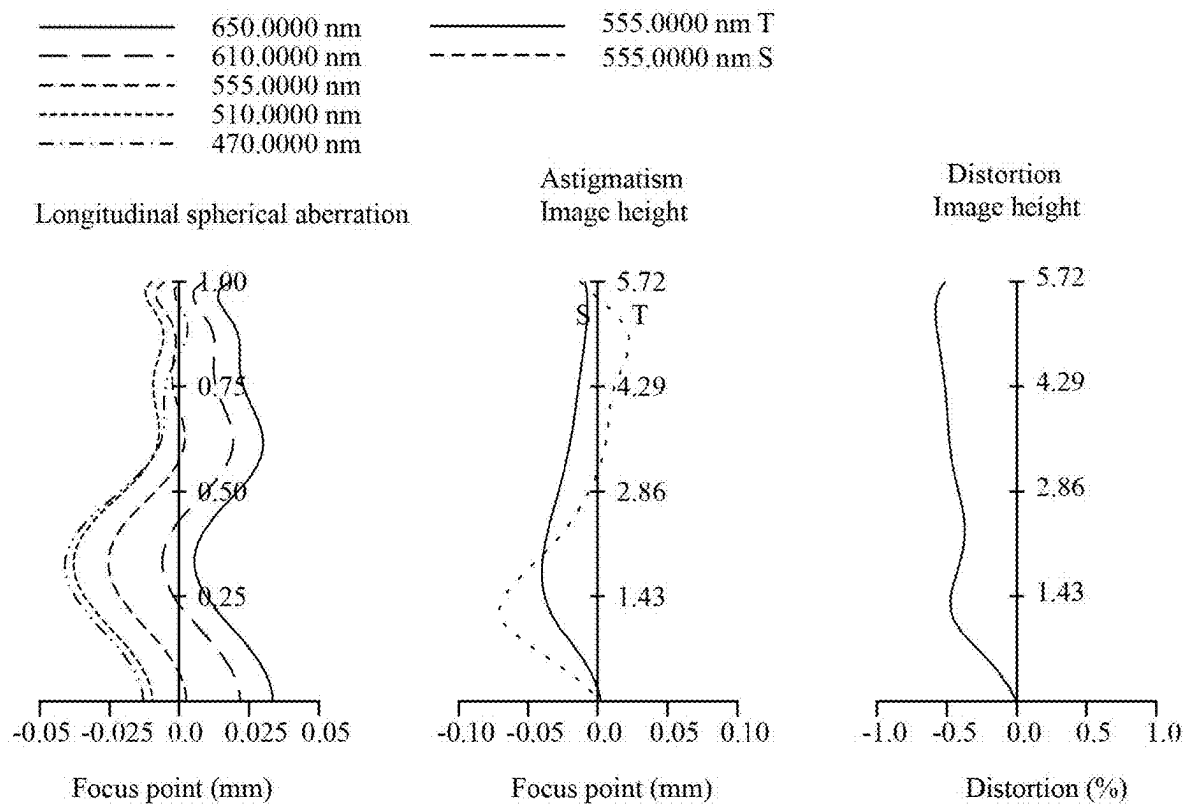


FIG. 5B

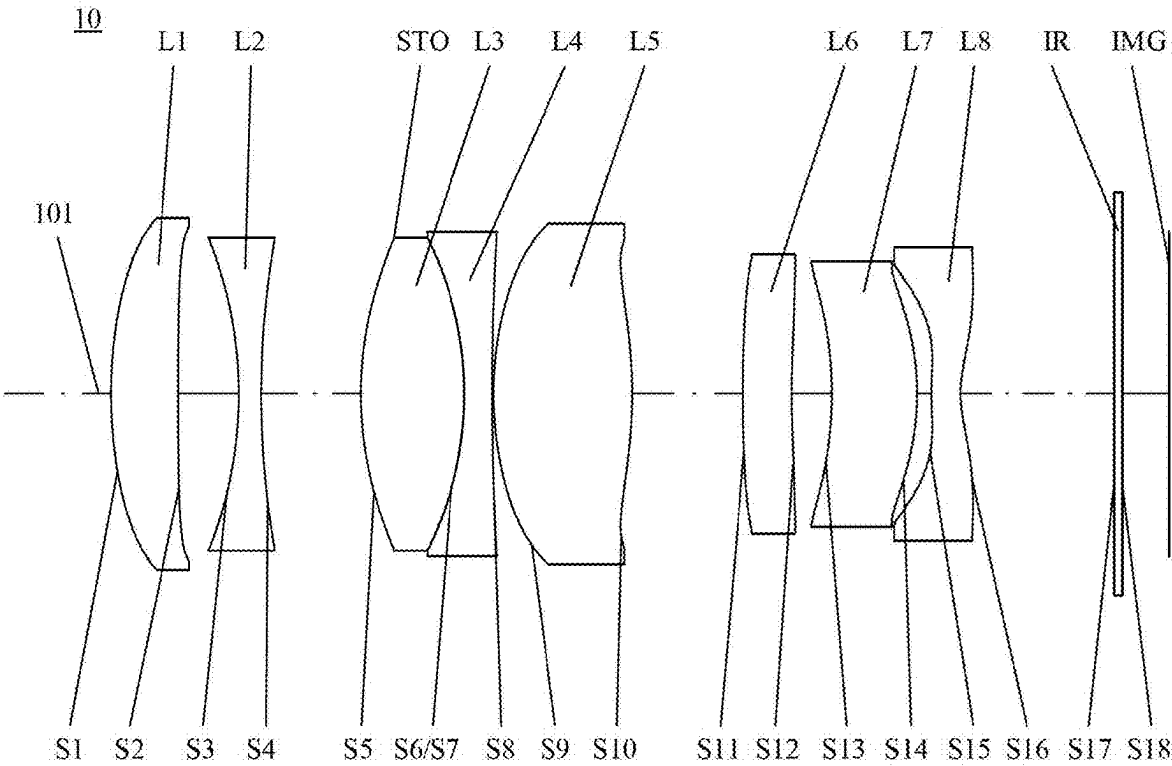


FIG. 5C

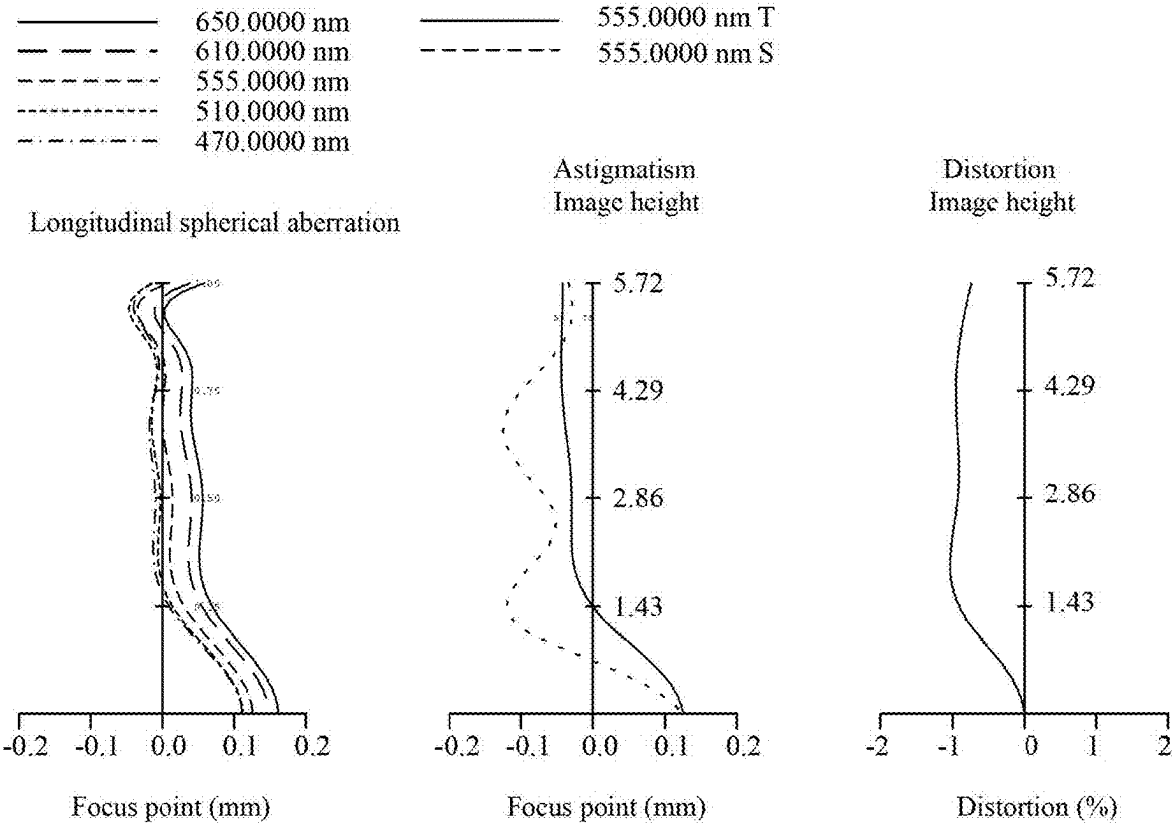


FIG. 5D

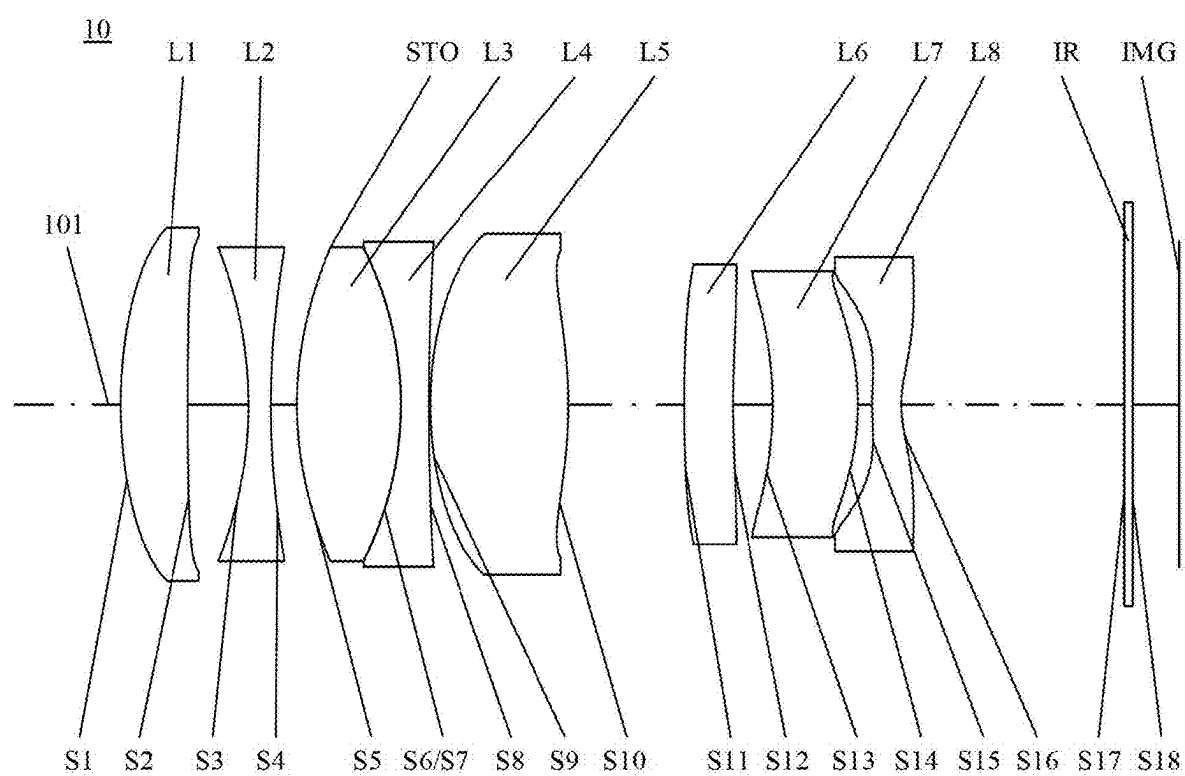
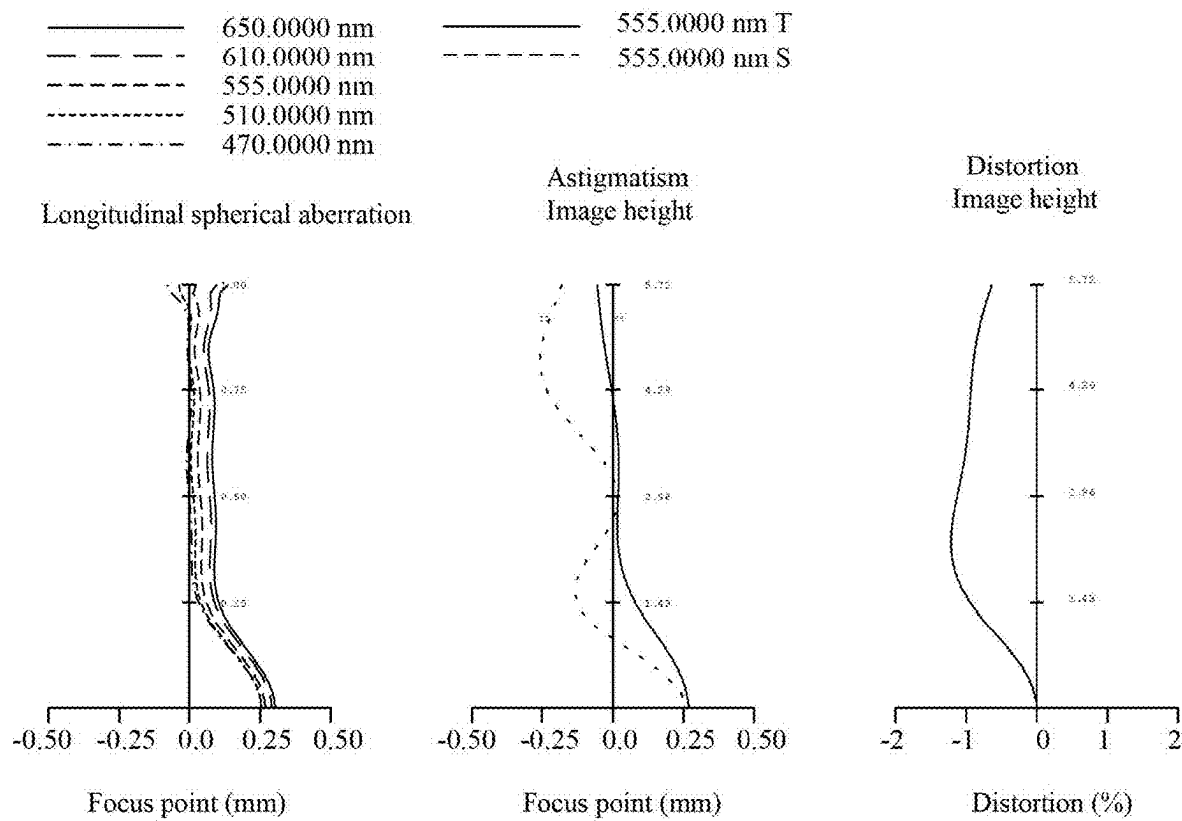


FIG. 5E



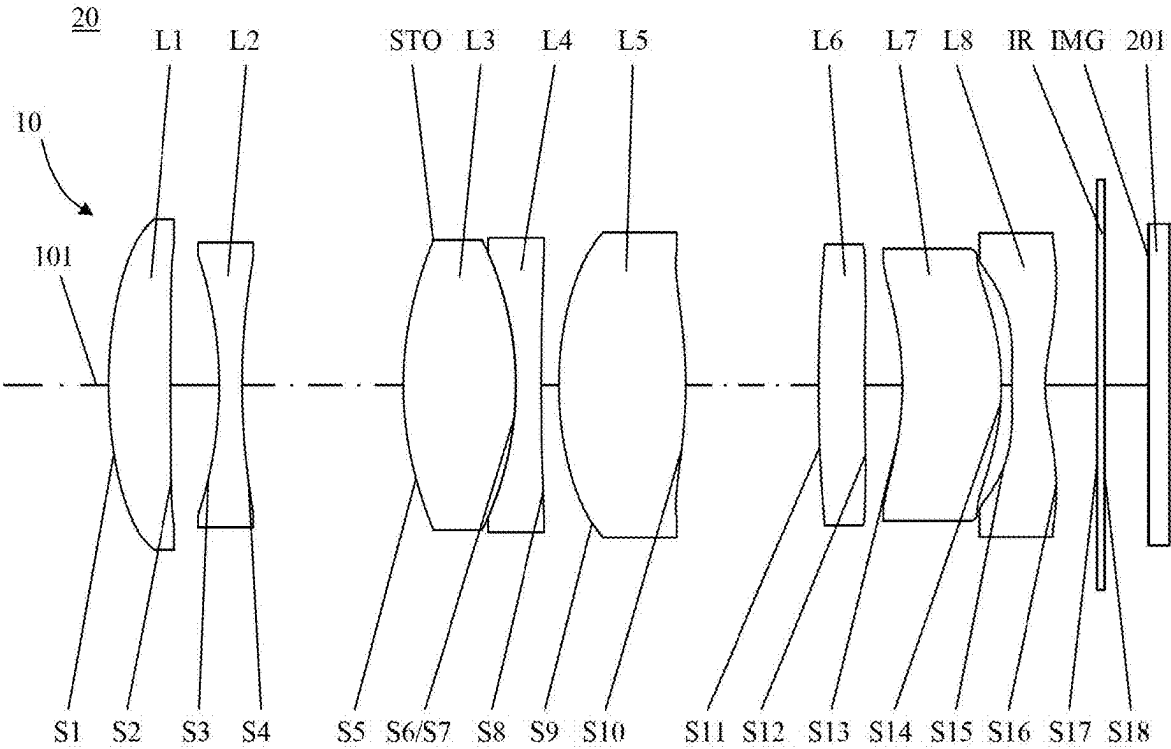


FIG. 6

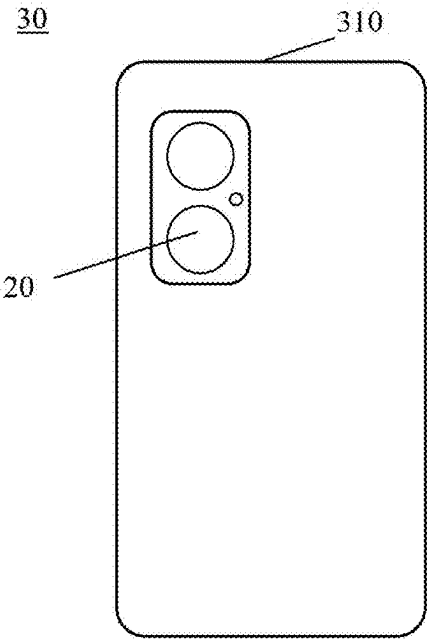


FIG. 7

OPTICAL SYSTEM, CAMERA MODULE, AND ELECTRONIC DEVICE

FIELD

[0001] The subject matter relates to field of optical imaging, and more particularly, to an optical system, a camera module, and an electronic device.

BACKGROUND

[0002] With the development of technologies, especially optical systems technologies, optical systems are widely used in smart portable devices such as mobile phones and tablet computers, and consumers have increased requirements for focal length coverage in the optical systems. However, the imaging quality of existing optical systems with zooming capability is poor and cannot meet the increased requirements. How to improve the imaging quality of the optical system with zooming capability is thus problematic.

SUMMARY

[0003] The present disclosure discloses an optical system, a camera module, and an electronic device with zooming capability, being miniaturized, and with imaging quality are needed.

[0004] The present disclosure provides the following embodiments.

[0005] A first aspect of the present disclosure provides an optical system consisting of eight lenses with refractive power. From an object side to an image side along an optical axis of the optical system, the eight lenses sequentially include a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens. The first lens and the second lens form a first lens group with negative refractive power. The third lens to the fifth lens form a second lens group with positive refractive power. The sixth lens to the eighth lens form a third lens group having negative refractive power. The first lens group is fixed relative to an imaging plane of the optical system. Each of the second lens group and the third lens group is configured to move along the optical axis between the first lens group and the imaging plane to realize zooming capability of the optical system. The optical system satisfies relationships of $1 < f_{\max}/f_{\min} < 2.1$, $-1.5 < f_{12}/f_{\max} < -0.5$, $0.25 < f_{345}/f_{\max} < 0.8$, and $-1.1 < f_{678}/f_{\max} < -0.3$. Wherein, $f_{\max}$ is a maximum focal length of the optical system, $f_{\min}$ is a minimum focal length of the optical system, f_{12} is a combined focal length of the first lens to the second lens, f_{345} is a combined focal length of the third lens to the fifth lens, and f_{678} is a combined focal length of the sixth lens to the eighth lens.

[0006] The present disclosure moves the second lens group and the third lens group between the first lens group and the imaging plane, thereby enabling the optical system to realize zooming capability.

[0007] The total length of the optical system does not change during zooming. That is, the optical system has a fixed total length, which meets the miniaturization requirements of the optical system. The eight lenses with refractive power evenly distribute the burden of light refraction to each lens, thereby reducing the workload of a single lens for bending light and avoiding excessive bending of the lens that will increase tolerance sensitivity. Moreover, since each

of the second lens group and the third lens group has three lenses, the two lens groups have sufficient adjustment capability to ensure clear imaging. Also, too many lenses in a single lens group are avoided, which reduces the momentum and power provided by a motor during zooming, thereby reducing the burden of the optical system onto the motor and achieving fast response. The reasonable distribution of refractive power balances aberrations and improves the imaging quality of the optical system. The reasonable surface shapes make the lenses more compatible, which effectively utilizes the space, smoothly propagates the light, reduces the introduction of aberrations, and improves the imaging quality.

[0008] A second aspect of the present disclosure further provides a camera module, which includes the optical system as described in any embodiment of the first aspect and a photosensitive chip. The photosensitive chip is located on the image side of the optical system. The optical system of the present disclosure is incorporated into the camera module. By reasonably designing the surface shape and the refractive power of each lens in the optical system and the optical system to have a fixed total optical length, the camera module realizes zooming capability, is miniaturized, and has high imaging quality.

[0009] A third aspect of the present disclosure further provides an electronic device, which includes a housing and a camera module as described in the second aspect. The camera module is received in the housing. By incorporating the camera module of the present disclosure into the electronic device, the electronic device realizes zooming capability and has high imaging quality and fast response speed. The miniaturized camera module also saves more space for installing other components.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Implementations of the present technology will now be described, by way of embodiment, with reference to the attached figures. Obviously, the drawings are only some embodiments of the present disclosure. For those ordinary skill in the art, other drawings can be obtained based on these drawings without creative work.

[0011] FIG. 1A is a schematic diagram of an optical system according to a first embodiment of the present disclosure in a short focal state.

[0012] FIG. 1B shows an aberration diagram of the optical system of the first embodiment in the short focal state.

[0013] FIG. 1C is a schematic diagram of the optical system of the first embodiment in a medium telephoto state.

[0014] FIG. 1D shows an aberration diagram of the optical system of the first embodiment in the medium telephoto state.

[0015] FIG. 1E is a schematic diagram of the optical system of the first embodiment in a telephoto state.

[0016] FIG. 1F shows an aberration diagram of the optical system of the first embodiment in the telephoto state.

[0017] FIG. 2A is a schematic diagram of an optical system according to a second embodiment of the present disclosure in a short focal state.

[0018] FIG. 2B shows an aberration diagram of the optical system of the second embodiment in the short focal state.

[0019] FIG. 2C is a schematic diagram of the optical system of the second embodiment in a medium telephoto state.

[0020] FIG. 2D shows an aberration diagram of the optical system of the second embodiment in the medium telephoto state.

[0021] FIG. 2E is a schematic diagram of the optical system according to the second embodiment in a telephoto state.

[0022] FIG. 2F shows an aberration diagram of the optical system of the second embodiment in the telephoto state.

[0023] FIG. 3A is a schematic diagram of an optical system according to a third embodiment of the present disclosure in a short focal state.

[0024] FIG. 3B shows an aberration diagram of the optical system of the third embodiment in the short focal state.

[0025] FIG. 3C is a schematic diagram of the optical system of the third embodiment in a medium telephoto state.

[0026] FIG. 3D shows an aberration diagram of the optical system of the third embodiment in the medium telephoto state.

[0027] FIG. 3E is a schematic diagram of the optical system of the third embodiment in a telephoto state.

[0028] FIG. 3F shows an aberration diagram of the optical system of the third embodiment in the telephoto state.

[0029] FIG. 4A is a schematic diagram of an optical system according to a fourth embodiment of the present disclosure in a short focal state.

[0030] FIG. 4B shows an aberration diagram of the optical system of the fourth embodiment in the short focal state.

[0031] FIG. 4C is a schematic diagram of the optical system of the fourth embodiment in a medium telephoto state.

[0032] FIG. 4D shows an aberration diagram of the optical system of the fourth embodiment in the medium telephoto state.

[0033] FIG. 4E is a schematic diagram of the optical system of the fourth embodiment in a telephoto state.

[0034] FIG. 4F shows an aberration diagram of the optical system of the fourth embodiment in the telephoto state.

[0035] FIG. 5A is a schematic diagram of an optical system according to a fifth embodiment of the present disclosure in a short focal state.

[0036] FIG. 5B shows an aberration diagram of the optical system of the fifth embodiment in the short focal state.

[0037] FIG. 5C is a schematic diagram of the optical system of the fifth embodiment in a medium telephoto state.

[0038] FIG. 5D shows an aberration diagram of the optical system of the fifth embodiment in the medium telephoto state.

[0039] FIG. 5E is a schematic diagram of the optical system of the fifth embodiment in a telephoto state.

[0040] FIG. 5F shows an aberration diagram of the optical system of the fifth embodiment in the telephoto state.

[0041] FIG. 6 is a schematic diagram of a camera module according to an embodiment of the present disclosure.

[0042] FIG. 7 is a schematic diagram of an electronic device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0043] Implementations of the present disclosure will now be described, by way of embodiments, with reference to the above figures. The embodiments are obviously a portion but not all of the embodiments of the present application. Based on the embodiments of the present application, other

embodiments obtained by ordinary skill in the art without creative work will still fall within the scope of protection of the present application.

[0044] A first aspect of the present disclosure provides an optical system consisting of eight lenses with refractive power. From an object side to an image side along an optical axis of the optical system, the eight lenses sequentially include a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens. The first lens and the second lens form a first lens group with negative refractive power. The third lens to the fifth lens form a second lens group with positive refractive power. The sixth lens to the eighth lens form a third lens group having negative refractive power. The first lens group is fixed relative to an imaging plane of the optical system. The second lens group and the third lens group are configured to move along the optical axis between the first lens group and the imaging plane to realize zooming capability of the optical system. The present disclosure moves the second lens group and the third lens group between the first lens group and the imaging plane, thereby enabling the optical system to realize zooming capability. The total length of the optical system does not change during zooming. That is, the optical system has a fixed total length, which meets the miniaturization requirements of the optical system. The eight lenses with refractive power evenly distribute the burden of light refraction to each lens, thereby reducing the workload of a single lens for bending light and avoiding excessive bending of the lens that will increase tolerance sensitivity. Moreover, since each of the second lens group and the third lens group includes three lenses, the two lens groups have sufficient adjustment capability to ensure clear imaging. Also, too many lenses in a single lens group are avoided, which reduces the momentum and power provided by a motor during zooming, thereby reducing the burden of the optical system onto the motor and achieving fast response. The reasonable distribution of refractive power balances aberrations and improves the imaging quality of the optical system. The reasonable surface shapes make the lenses more compatible, which effectively utilizes the space, smoothly propagates the light, reduces the introduction of aberrations, and improves the imaging quality.

[0045] In some embodiments, $1 < f_{\max}/f_{\min} < 2.1$. Wherein, $f_{\max}$ is a maximum focal length of the optical system, and $f_{\min}$ is a minimum focal length of the optical system. The above relationship ensures the zooming capability of the optical system and multiple focal length coverage. Furthermore, $1.1 < f_{\max}/f_{\min} < 1.9$.

[0046] In some embodiments, $-1.5 < f_{12}/f_{\max} < -0.5$. Wherein, f_{12} is a combined focal length of the first lens to the second lens. The above relationship allows the refractive power of the first lens group to be moderate, which avoids excessive bending of the lenses. Furthermore, $-1.3 < f_{12}/f_{\max} < -0.8$.

[0047] In some embodiments, $0.25 < f_{345}/f_{\max} < 0.8$. Wherein, f_{345} is a combined focal length of the third lens to the fifth lenses. The above relationship allows the refractive power of the second lens group to be moderate, which avoids excessive bending of the lenses and effectively adjusts the focal length of the optical system.

[0048] In some embodiments, $-1.1 < f_{678}/f_{\max} < -0.3$. Wherein, f_{678} is a combined focal length of the sixth lens to the eighth lenses. The above relationship allows the refractive power of the third lens group to be moderate,

which avoids excessive bending of the lenses, achieves efficient focusing, and avoids the introduction of too much aberrations, thereby improving the imaging quality.

[0049] In some embodiments, $1 < \text{FOV}_{\text{max}}/\text{FOV}_{\text{min}} < 2.1$. Wherein, FOV_{max} is a maximum field of view of the optical system, and FOV_{min} is a minimum field of view of the optical system. The above relationship ensures the wide variation of the field of view and improves the applicability of shooting scenarios. Furthermore, $1.1 < \text{FOV}_{\text{max}}/\text{FOV}_{\text{min}} < 1.9$.

[0050] In some embodiments, $30^\circ < \text{FOV}_{\text{max}} < 45^\circ$. The above relationship allows the optical system to exhibit wide-angle characteristics in the short focal state. Furthermore, $32^\circ < \text{FOV}_{\text{max}} < 42^\circ$.

[0051] In some embodiments, $16^\circ < \text{FOV}_{\text{min}} < 36^\circ$. The above relationship allows the optical system to exhibit telephoto characteristics in the telephoto state. Furthermore, $18^\circ < \text{FOV}_{\text{min}} < 34^\circ$.

[0052] In some embodiments, $1 < \text{FNO}_{\text{max}}/\text{FNO}_{\text{min}} < 1.8$. Wherein, FNO_{max} is a maximum F-number of the optical system, and FNO_{min} is a minimum F-number of the optical system. The above relationship regulates different light intensities and avoids significant changes in the brightness of the captured images caused by excessive changes in the aperture during zooming. Furthermore, $1.1 < \text{FNO}_{\text{max}}/\text{FNO}_{\text{min}} < 1.6$.

[0053] In some embodiments, $2.4 < \text{FNO}_{\text{max}} < 3.4$. The above relationship allows the optical system to control the amount of incoming light, avoids the introduction of excessive aberrations, and improves the depth of field. Furthermore, $2.6 < \text{FNO}_{\text{max}} < 3.2$.

[0054] In some embodiments, $1.5 < \text{FNO}_{\text{min}} < 2.6$. The above relationship allows the optical system to increase the amount of incoming light and achieves the shooting effect of background blurring. Furthermore, $1.7 < \text{FNO}_{\text{min}} < 2.4$.

[0055] In some embodiments, $-1.9 < f_{12}/f_{345} < -2.7$. The above relationship allows the refractive power of the first lens group and the second lens group to be compatible with each other, which balances aberrations, reducing system sensitivity, and improves zooming efficiency, thereby enhancing the overall resolution and response speed of the optical system. Furthermore, $-2.1 < f_{12}/f_{345} < -2.5$.

[0056] In some embodiments, $1.2 < f_{12}/f_{678} < 2.1$. The above relationship allows the refractive power of the first lens group and the third lens group to be compatible with each other, which balances aberrations, reduces system sensitivity, and improves zooming efficiency, thereby enhancing the overall resolution and response speed of the optical system. Furthermore, $1.4 < f_{12}/f_{678} < 1.95$.

[0057] In some embodiments, $5 < \text{TTL}/\text{ImgH} < 9$. Wherein, TTL is a distance from an object side surface of the first lens to the imaging plane at the optical axis, i.e. the total optical length, and ImgH is half of an image height corresponding to a maximum field of view of the optical system. The above relationship enables the optical system to have telephoto and miniaturized characteristics, thereby facilitating the placement of a camera module containing the optical system into an electronic device. Furthermore, $6 < \text{TTL}/\text{ImgH} < 8$.

[0058] In some embodiments, $0.3 < \text{Yc82}/\text{SD82} < 0.9$. Wherein, Yc82 is a vertical height from an off-axis vertex of an image side surface of the eighth lens to the optical axis, and SD82 is an effective half-aperture of the image side surface of the eighth lens. By satisfying the above relationship, the travelling direction of the light is adjusted, which

improves the imaging quality such as relative illumination at the margin area of the imaging plane. Also, the surface shape of the eighth lens is adjusted to correct off-axis aberrations such as curvature of field. The incident angle of light at the imaging surface is also adjusted to enhance the response efficiency of the electronic photosensitive component, thereby further improving the imaging quality such as relative illumination.

[0059] It should be noted that the off-axis vertex refers to an off-axis point of the surface curve at which the first-order derivative is zero. Referring to FIG. 1, which is a schematic diagram showing the parameter Yc82 and the corresponding off-axis vertex E according to the first embodiment of the present disclosure. Furthermore, $0.5 < \text{Yc82}/\text{SD82} < 0.8$.

[0060] In some embodiments, $0.8 < \text{SD}_{\text{max}}/\text{ImgH} < 1.3$. Wherein, SD_{max} is a maximum effective half-aperture from the object side surface of the first lens to the image side surface of the eighth lens. The above relationship ensures that the aperture of the optical system is substantially equal to the size of the imaging surface, which helps to unify the radial size, improves the space utilization, and achieves miniaturization. Furthermore, $0.9 < \text{SD}_{\text{max}}/\text{ImgH} < 1.2$.

[0061] In some embodiments, $1.5 < \text{GL2}/\text{GL1} < 2.7$. Wherein, GL1 is a distance between the object side surface of the first lens and the image side surface of the second lens at the optical axis, and GL2 is a distance between an object side surface of the third lens and an image side surface of the fifth lens at the optical axis. The above relationship helps to reasonably allocate the axial space and leave sufficient stroke space for zooming. Furthermore, $1.7 < \text{GL2}/\text{GL1} < 2.5$.

[0062] In some embodiments, $0.9 < \text{GL2}/\text{GL3} < 1.8$. Wherein, GL3 is a distance between an object side surface of the sixth lens and the image side surface of the eighth lens at the optical axis. The above relationship allows the sizes of the second lens group and the third lens group to match each other and the imaging performances to be similar, which balances the aberrations and improves the imaging quality. Furthermore, $1 < \text{GL2}/\text{GL3} < 1.65$.

[0063] In some embodiments, $1 < \text{TD}_{\text{max}}/\text{TD}_{\text{min}} < 1.5$. Wherein, TD_{max} is a maximum distance between the object side surface of the first lens and the image side surface of the eighth lens at the optical axis, and TD_{min} is a minimum distance between the object side surface of the first lens and the image side surface of the eighth lens at the optical axis. By satisfying the above relationship, the zooming stroke length is moderate, which reduces the response time for zooming and provides sufficient zooming range. Furthermore, $1 < \text{TD}_{\text{max}}/\text{TD}_{\text{min}} < 1.3$.

[0064] In some embodiments, $0.5 < \Delta A/\Delta C < 1.1$. Wherein, ΔA is a maximum stroke of the second lens group at the optical axis, and ΔC is a maximum stroke of the third lens group at the optical axis. The above relationship allows the strokes of the two lens groups to be close with more overlapping space, which saves the space and improves the zooming capability. Furthermore, $0.6 < \Delta A/\Delta C < 0.9$.

[0065] In some embodiments, the first lens has positive refractive power, which shortens the total length of the optical system. The object side surface of the first lens is convex near the optical axis, thereby reducing the incident angle of light at the object side surface and improving the field of view. The second lens has negative refractive power, thereby eliminating the negative spherical aberration generated by the first lens. The object side surface of the second lens is concave near the optical axis, and the image side

surface of the second lens is concave near the optical axis, thereby improving the negative refractive power of the second lens and preventing the incident light from converging too quickly. The third lens has positive refractive power, and when the third lens is combined with the fourth lens, the system chromatic aberrations are eliminated. The object side surface of the third lens is convex near the optical axis, and the image side surface of the third lens is convex near the optical axis, thereby matching the surface shapes of the fourth lens and eliminate aberrations. The fourth lens has negative refractive power, and when the fourth lens is combined with the third lens, the system chromatic aberrations are eliminated. The object side surface of the fourth lens is concave near the optical axis, thereby matching the surface shape of the third lens and eliminates aberrations. The fifth lens has positive refractive power, thereby converging the light and preventing the aperture of the sixth lens from being too large. The object side surface of the fifth lens is convex near the optical axis, and the image side surface of the fifth lens is convex near the optical axis, thereby improving the refractive power of the fifth lens. The sixth lens has refractive power. The object side surface of the sixth lens is convex near the optical axis, and the image side surface of the sixth lens is concave near the optical axis. When the sixth lens is combined with the seventh lens, the aberrations are corrected. The seventh lens has refractive power. The object side surface of the seventh lens is concave near the optical axis, and the image side surface of the seventh lens is convex near the optical axis. When the seventh lens is combined with the sixth lens, the aberrations are corrected. The eighth lens has negative refractive power, thereby correcting axial aberrations. The object side surface of the eighth lens is convex near the optical axis, and the image side surface of the eighth lens is concave near the optical axis, thereby reducing the incident angle of light near the optical axis at the surfaces of the eighth lens and improving the transmissivity.

[0066] In some embodiments, the third lens and the fourth lens are cemented to eliminate the system chromatic aberrations.

[0067] In some embodiments, $0.7 < f_1/f_{\max} < 1.6$. Wherein, f_1 is a focal length of the first lens. By satisfying the above relationship, the refractive power is reasonably distributed, which improves the imaging quality. Furthermore, $0.9 < f_1/f_{\max} < 1.55$.

[0068] In some embodiments, $-0.8 < f_2/f_{\max} < -0.2$. Wherein, f_2 is a focal length of the second lens. By satisfying the above relationship, the refractive power is reasonably distributed, which improves the imaging quality. Furthermore, $-0.7 < f_2/f_{\max} < -0.3$.

[0069] In some embodiments, $0.2 < f_3/f_{\max} < 0.6$. Wherein, f_3 is a focal length of the third lens. By satisfying the above relationship, the refractive power is reasonably distributed, which improves the imaging quality. Furthermore, $0.25 < f_3/f_{\max} < 0.55$.

[0070] In some embodiments, $-0.8 < f_4/f_{\max} < -0.2$. Wherein, f_4 is a focal length of the fourth lens. By satisfying the above relationship, the refractive power is reasonably distributed, which improves the imaging quality. Furthermore, $-0.7 < f_4/f_{\max} < -0.3$.

[0071] In some embodiments, $0.25 < f_5/f_{\max} < 1$. Wherein, f_5 is a focal length of the fifth lens. By satisfying the above

relationship, the refractive power is reasonably distributed, which improves the imaging quality. Furthermore, $0.4 < f_5/f_{\max} < 0.85$.

[0072] In some embodiments, $1.2 < f_8/f_{\max} < -0.25$. Wherein, f_8 is a focal length of the sixth lens. By satisfying the above relationship, the refractive power is reasonably distributed, which improves the imaging quality. Furthermore, $-1 < f_8/f_{\max} < -0.3$.

[0073] In some embodiments, $0.4 < R_1/f_{\max}$. Wherein, R_1 is a radius of curvature of the object side surface of the first lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $0.5 < R_1/f_{\max} < 3.5$.

[0074] In some embodiments, $0.8 < R_4/f_{\max}$. Wherein, R_4 is a radius of curvature of the image side surface of the second lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $0.8 < R_4/f_{\max} < 5$.

[0075] In some embodiments, $0.3 < R_5/f_{\max}$. Wherein, R_5 is a radius of curvature of the object side of the third lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $0.3 < R_5/f_{\max} < 2.5$.

[0076] In some embodiments, $-0.6 < R_6/f_{\max} < -0.2$. Wherein, R_6 is a radius of curvature of the image side surface of the third lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $-0.5 < R_6/f_{\max} < -0.3$.

[0077] In some embodiments, $0.3 < R_9/f_{\max} < 1$. Wherein, R_9 is a radius of curvature of the object side surface of the fifth lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $0.35 < R_9/f_{\max} < 0.9$.

[0078] In some embodiments, $-1 < R_{10}/f_{\max} < -0.4$. Wherein, R_{10} is a radius of curvature of the image side surface of the fifth lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $-0.95 < R_{10}/f_{\max} < -0.4$.

[0079] In some embodiments, $1.5 < R_{11}/f_{\max}$. Wherein, R_{11} is a radius of curvature of the object side surface of the sixth lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $2 < R_{11}/f_{\max} < 4$.

[0080] In some embodiments, $1 < R_{12}/f_{\max}$. Wherein, R_{12} is a radius of curvature of the image side surface of the sixth lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $1.3 < R_{12}/f_{\max} < 4$.

[0081] In some embodiments, $-0.9 < R_{13}/f_{\max} < -0.3$. Wherein, R_{13} is a radius of curvature of the object side surface of the seventh lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $-0.7 < R_{13}/f_{\max} < -0.35$.

[0082] In some embodiments, $-0.9 < R_{14}/f_{\max} < -0.2$. Wherein, R_{14} is a radius of curvature of the image side surface of the seventh lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $-0.75 < R_{14}/f_{\max} < -0.3$.

[0083] In some embodiments, $0.2 < R15/f_{\max} < 1.3$. Wherein, R15 is a radius of curvature of the object side surface of the eighth lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $0.25 < R15/f_{\max} < 1.2$.

[0084] In some embodiments, $0.1 < R16/f_{\max} < 0.4$. Wherein, R16 is a radius of curvature of the image side surface of the eighth lens at the optical axis. By satisfying the above relationship, the surface shape is reasonably designed, which improves the imaging quality. Furthermore, $0 < R16/f_{\max} < 0.4$.

[0085] In some embodiments, $1.4 < CT1/CT2 < 3.5$. Wherein, CT1 is a thickness of the first lens at the optical axis, and CT2 is a thickness of the second lens at the optical axis. The above relationship balances the axial chromatic aberrations generated by the first lens. Furthermore, $1.8 < CT1/CT2 < 3.2$.

[0086] In some embodiments, $3 < CT3/CT4 < 5$. Wherein, CT3 is a thickness of the third lens at the optical axis, and CT4 is a thickness of the fourth lens at the optical axis. The above relationship allows the third lens and the fourth lens to match each other, which eliminates the on-axis chromatic aberrations. Furthermore, $3.2 < CT3/CT4 < 4.7$.

[0087] In some embodiments, $4 < CT5/CT4 < 5.5$. Wherein, CT5 is a thickness of the fifth lens at the optical axis. The above relationship prevents the fifth lens from being too thin, thereby ensuring a sufficient light deflection space in the fifth lens. Furthermore, $4.4 < CT5/CT4 < 5.2$.

[0088] In some embodiments, $1.1 < CT7/CT6 < 2.8$. Wherein, CT6 is a thickness of the sixth lens at the optical axis, and CT7 is a thickness of the seventh lens at the optical axis. The above relationships allow the thicknesses of the sixth and seventh lenses to be similar, thereby achieving even distribution of refractive power. Furthermore, $1.5 < CT7/CT6 < 2.4$.

[0089] In some embodiments, $2.2 < CT7/CT8 < 3.7$. Wherein, CT8 is a thickness of the eighth lens at the optical axis. The above relationship allows the thickness of the eighth lens to be moderate, which eliminates the off-axis chromatic aberrations. Furthermore, $2.6 < CT7/CT8 < 3.2$.

[0090] The above technical features of the optical system of the present disclosure may be combined with each other to achieve corresponding effects.

[0091] In one embodiment, the lenses may be made of glass or plastic. When the lenses are made of glass, the degree of freedom in designing the refractive power of the lenses of the optical system is increased, and the glass lenses may be made by grinding or mechanical molding. When the lenses are made of plastic, the manufacturing cost is reduced. In addition, spherical or aspherical surface (ASPs) may be included at the surfaces of the lenses. The spherical lenses reduce the manufacturing difficulty. When the lenses have aspherical surfaces, more control variables are obtained to reduce aberrations, reduce the number of lenses, and reduce the total length of the optical system of the present disclosure. The aspherical lenses may be made by injection molding or mechanical molding.

[0092] In one embodiment, additives may be selectively added to at least one lens material to produce light absorption or interference effects. The additives change the trans-

missivity of the lens to specific wavelengths of light, thereby reducing stray light and color deviation. Optionally, when the lens has an aspherical surface, it means that the entire or a portion of the optical effective area of the lens is aspherical.

[0093] In one embodiment, when the surface of the lens is convex but the position of the convex surface is not defined, it means that the surface of the lens may be convex near the optical axis. When the surface of the lens is concave but the position of the concave surface is not defined, it means that the surface of the lens may be concave near the optical axis. In the optical system of the present disclosure, when the lens is indicated to have positive or negative refractive power or have a focal length, the refractive power or focal length is near the optical axis. Optionally, a critical point is a tangent point on the surface of the lens, which is tangent to a plane perpendicular to the optical axis, except for the intersection point with the optical axis. The inflection point is an intersection point on the surface of the lens, which changes from a positive curvature to a negative curvature.

[0094] In one embodiment, the optical system may further include at least one component with light refraction function, such as a prism or a reflector, which is selectively installed on the optical path between the target subject and the imaging plane. The component is used to provide a more flexible spatial configuration for the optical system, such that the miniaturization of the electronic device is not limited by the total optical length of the optical system. Optionally, the optical system may selectively include three or more light refraction components, and the types, the quantities, and positions of the light refraction components are not limited.

[0095] Implementations of the present technology will now be described with reference to different embodiments and figures.

First Embodiment

[0096] Referring to FIGS. 1A to 1F, the optical system 10 of the embodiment consists of following eight lenses from the object side to the image side.

[0097] The first lens L1 has positive refractive power. Near the optical axis, the object side surface S1 is convex, and the image side surface S2 is concave.

[0098] The second lens L2 has negative refractive power. Near the optical axis, the object side surface S3 is concave, and the image side surface S4 is concave.

[0099] The third lens L3 has positive refractive power. Near the optical axis, the object side surface S5 is convex, and the image side surface S6 is convex.

[0100] The fourth lens L4 has negative refractive power. Near the optical axis, the object side surface S7 is concave, and the image side surface S8 is concave.

[0101] The fifth lens L5 has positive refractive power. Near the optical axis, the object side surface S9 is convex, and the image side surface S10 is convex.

[0102] The sixth lens L6 has positive refractive power. Near the optical axis, the object side surface S11 is convex, and the image side surface S12 is concave.

[0103] The seventh lens L7 has positive refractive power. Near the optical axis, the object side surface S13 is concave, and the image side surface S14 is convex.

[0104] The eighth lens L8 has negative refractive power. Near the optical axis, the object side surface S14 is convex, and the image side surface S16 is concave.

[0105] The first lens L1 to the eighth lens L8 include both glass lenses and plastic lenses. In other embodiments, all the lenses may also be glass lenses or plastic lenses. The third lens L3 and the fourth lens L4 are cemented. In other embodiments, other lenses or none of the lenses may be cemented.

[0106] The optical system 10 further includes an aperture stop STO and a filter IR. In the embodiment, the aperture stop STO is located at the object side surface of the third lens L3. In other embodiments, the aperture stop STO may also be separated from the object side surface of the third lens L3. For example, the aperture stop STO may be located between the first lens L1 and the second lens L2 or on the object side of the first lens L1. In the embodiment, the filter IR is an infrared cutoff filter made of glass with a coating thereon. The filter IR is located between the eighth lens L8 and the imaging plane IMG, and includes an object side surface S17 and an image side surface S18. In other embodiments, the filter IR is not limited to infrared cutoff filters, but may also be a dual channel filter, an infrared filter, or other types of filters. The filter IR may also be made of plastic.

[0107] The imaging plane IMG of the optical system 10 is located at the image side of the optical system 10. When the optical system 10 is mounted into the camera module, the imaging plane IMG overlaps with the effective pixel area of the photosensitive chip.

[0108] The first lens L1 to the second lens L2 of the optical system 10 are fixed to each other to form the first lens group.

plane IMG. Each of the second lens group and the third lens group is able to move along the optical axis 101 between the first lens group and the imaging plane IMG, thereby achieving the zooming capability of the optical system 10. In the embodiment, the filter IR is fixed relative to the imaging plane IMG, which reduces the weight of the moving lens groups and improves the zooming speed. In other embodiments, the filter IR may also be fixed to the third lens group or the second lens group for facilitating the assembly of the lenses.

[0109] Table 1A shows detailed structural parameters of the optical system 10 of the embodiment. The surface numerals S1 and S2 are the object side surface S1 and the image side surface S2 of the first lens L1, respectively. For a specific lens, the surface with a smaller surface numeral is the object side surface, and the surface with a larger surface numeral is the image side surface, and so forth for other lenses. The focal length, the refractive index, and Abbe number are obtained by visible rays with a reference wavelength of 555 nm. The Y radius is the radius of curvature of the object side surface or image side surface of the related surface numeral at the optical axis 101. In the “thickness” column of the first lens L1, the first value is the thickness of the lens at the optical axis 101, and the second value is the distance from the image side surface of the lens to the next optical surface (i.e., the object side surface of the next lens or the surface of the aperture stop) at the optical axis 101, and so forth for other lenses. The units of the Y radius, the thickness, and the focal length are all millimeters (mm).

TABLE 1A

Surface numeral	Name	Surface type	Y radius (mm)	Thickness (mm)	Material	Refractive index	Abbe number	Focal length (mm)
	Object side	sphere	infinity	infinity				
S1	first lens L1	asphere	17.583	2.411	plastic	1.588	28.400	30.292
S2		asphere	916.263	1.867				
S3	second lens L2	sphere	-15.834	0.878	glass	1.800	46.500	-12.460
S4		sphere	28.248	A				
S5(STO)	third lens L3	asphere	13.385	4.349	glass	1.678	55.600	10.285
S6	cemented surface	sphere	-12.725	0.005	cement			
S7	fourth lens L4	sphere	-12.725	1.000	glass	1.850	30.000	-13.449
S8		sphere	124.005	0.674				
S9	fifth lens L5	asphere	13.409	4.875	plastic	1.544	56.000	14.936
S10		asphere	-18.136	B				
S11	sixth lens L6	sphere	93.145	1.768	plastic	1.544	56.000	1147.374
S12		asphere	108.683	1.448				
S13	seventh lens L7	asphere	-12.046	3.800	plastic	1.670	19.200	93.152
S14		asphere	-11.404	0.403				
S15	eighth lens L8	asphere	18.799	1.312	plastic	1.544	56.000	-15.607
S16		asphere	5.719	C				
S17	filter IR	sphere	infinity	0.300	glass	1.510	64.000	—
S18		sphere	infinity	1.662				
IMG	imaging plane	sphere	infinity	0.000				

The third lens L3 to the fifth lens L5 are fixed to each other to form the second lens group. The sixth lens L6 to the eighth lens L8 are fixed to each other to form the third lens group. The first lens group is fixed relative to the imaging

[0110] Table 1B includes supplementary parameters to Table 1A, specifically the parameters when the optical system 10 is switched to the short focal state (Z1), the medium telephoto state (Z2), and the telephoto state (Z3).

TABLE 1B

	FNO	TTL (mm)	FOV (°)	ImgH (mm)	F (mm)	A (mm)	B (mm)	C (mm)
Short focal state (Z1)	2.200	40.100	34.300	5.720	18.700	6.203	5.134	1.991
Medium telephoto state (Z2)	2.600	40.100	26.000	5.720	24.800	3.530	4.207	5.592
Telephoto state (Z3)	3.000	40.100	21.000	5.720	30.500	1.262	4.334	7.730

[0111] Wherein, FNO is the F-number of the optical system **10**, TTL is the distance from the object side surface **S1** of the first lens **L1** to the imaging plane **IMG** of the optical system **10** at the optical axis **101**, FOV is the maximum field of view of the optical system, ImgH is half of the image height corresponding to the maximum field of view of the optical system **10**, f is the focal length of the optical system **10**, A is the distance between the first lens group and the second lens group at the optical axis **101**, B is the distance between the second lens group and the third lens group at the optical axis **101**, and C is the distance between the third lens group and the filter **IR** at the optical axis **101**.

[0112] According to Table 1B, when the optical system changes from the short focal state **Z1** to the telephoto state **Z3**, each of the second lens group and the third lens group moves along the optical axis **101** towards the first lens group. When the optical system changes from the telephoto state **Z3** to the short focal state **Z1**, each of the second lens group and the third lens group moves along the optical axis towards the imaging plane **IMG**.

[0113] It should be noted that Table 1B only provides exemplary parameters of the optical system **10** in the three zooming states. The switching between two zooming states

in the embodiment is continuous, and the optical system **10** also forms images during the switching process.

[0114] In the embodiment, a portion of the lenses have aspherical surfaces, and another portion of the lenses have spherical surfaces. In other embodiments, the surface shapes of the first lens **L1** to the eighth lens **L8** may all be spherical or aspherical. The surface shape of the aspherical surface in the embodiment may be expressed but not limited to the following aspherical formula:

$$x = \frac{ch^2}{1 + \sqrt{1 - (k+1)c^2h^2}} + \sum A_i h^i$$

[0115] Wherein, x is a distance from the corresponding point of the aspherical surface to a plane tangent to the on-axis vertex, h is a distance from the corresponding point of the aspherical surface to the optical axis **101**, c is a curvature of a vertex of the aspherical surface, k is cone constant, and A_i is a coefficient corresponding to the i^{th} high-order term in the above aspherical surface formula. Table 1C shows the high-order coefficients A4, A6, A8, A10, A12, A14, A16, A18 and A20 that applied for the aspherical surfaces in the first embodiment.

TABLE 1C

Surface numeral	k	A4	A6	A8	A10
S1	1.592287E-01	7.986195E-05	2.926520E-06	-3.050614E-07	4.098484E-08
S2	1.163521E-14	2.694112E-05	5.327606E-07	1.091634E-07	-4.503213E-10
S5	0.000000E+00	-4.084485E-05	-9.394938E-07	0.000000E+00	0.000000E+00
S9	-7.670579E+00	4.295773E-04	-3.809739E-06	4.684511E-07	-2.975808E-08
S10	-1.829882E+01	-3.179949E-05	1.118532E-05	-2.991184E-08	-9.234545E-09
S11	9.897088E+01	3.838071E-04	-2.630387E-05	6.264432E-07	5.718901E-09
S12	-9.885069E+01	3.283600E-04	-5.103610E-05	1.000583E-07	1.100605E-07
S13	3.558734E+00	8.749300E-04	-1.114450E-05	-6.225487E-06	9.948047E-07
S14	3.086159E+00	-3.575993E-03	1.221914E-03	-2.168562E-04	2.545918E-05
S15	-9.635550E+01	-1.290170E-02	2.224790E-03	-3.177599E-04	3.140763E-05
S16	-7.910389E+00	-6.017775E-03	8.386106E-04	-9.736914E-05	8.102243E-06
Surface numeral	A12	A14	A16	A18	A20
S1	-2.817539E-09	1.164068E-10	-2.738514E-12	3.337289E-14	-1.487201E-16
S2	-1.475607E-10	-2.789067E-12	1.104597E-12	-4.583855E-14	6.076087E-16
S5	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S9	1.903696E-09	-8.401448E-11	2.431581E-12	-4.007099E-14	2.819785E-16
S10	1.621438E-09	-1.034563E-10	3.886074E-12	-7.675170E-14	6.729287E-16
S11	-3.110621E-10	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S12	-2.588948E-09	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S13	-7.112565E-08	3.412947E-09	-1.093789E-10	2.070624E-12	-1.760318E-14
S14	-2.023127E-06	1.085753E-07	-3.771718E-09	7.647399E-11	-6.834205E-13
S15	-2.099957E-06	9.394179E-08	-2.731544E-09	4.746169E-11	-3.771880E-13
S16	-4.580670E-07	1.701185E-08	-3.951413E-10	5.180035E-12	-2.908376E-14

[0116] FIGS. 1B, 1D, and 1F show the aberration diagrams of the optical system **10** in the short focal state **Z1**, the medium telephoto state **Z2**, and the telephoto state **Z3**, respectively.

[0117] FIG. 1B shows the longitudinal spherical aberration diagram of the optical system **10** of the first embodiment in the short focal state **Z1**, FIG. 1D shows the longitudinal spherical aberration diagram of the optical system **10** of the first embodiment in the medium telephoto state **Z2**, and FIG. 1F shows the longitudinal spherical aberration diagram of the optical system **10** of the first embodiment in the telephoto state **Z3**. The reference wavelengths are 650 nm, 610 nm, 555 nm, 510 nm, 470 nm, and 435 nm. The horizontal coordinate along the X-axis represents the deviation of the focus point. The vertical coordinate along the Y-axis represents the normalized field of view. The longitudinal spherical aberration diagram represents the deviation of the focus point when the light with different wavelengths travels through the lenses of the optical system **10**. From the figures, the spherical aberration data of the optical system **10** in the first embodiment is good, indicating that the imaging quality of the optical system **10** in the embodiment is good.

[0118] FIG. 1B further shows the longitudinal astigmatism diagram of the optical system **10** of the first embodiment in the short focal state **Z2**, FIG. 1D further shows the longitudinal astigmatism diagram of the optical system **10** of the first embodiment in the medium telephoto state **Z2**, and FIG. 1F further shows the longitudinal astigmatism diagram of the optical system **10** of the first embodiment in the telephoto state **Z3**. The horizontal coordinate along the X-axis represents the deviation of the focus point. The vertical coordinate along the Y-axis represents the image height in

mm. The T curve of the astigmatism diagram represents the curvature of the imaging plane IMG in the tangential direction. The S curve of the astigmatism diagram represents the curvature of the imaging plane IMG in the sagittal direction. From the figures, astigmatism of the optical system **10** has been compensated.

[0119] FIG. 1B further shows the longitudinal distortion diagram of the optical system **10** of the first embodiment in the short focal state **Z1**, FIG. 1D further shows the longitudinal distortion diagram of the optical system **10** of the first embodiment in the medium telephoto state **Z2**, and FIG. 1F further shows the longitudinal distortion diagram of the optical system **10** of the first embodiment in the telephoto state **Z3**. The horizontal coordinate along the X-axis represents the distortion (%). The vertical coordinate along the Y-axis represents the image height. The distortion curve represents the magnitude of distortion corresponding to different field of views. From the figures, the distortion of the optical system **10** has been well corrected at the provided wavelengths.

[0120] Therefore, the optical system **10** of the embodiment has small aberrations and good imaging quality.

Second Embodiment

[0121] Referring to FIGS. 2A to 2F, the differences between the structures of the second embodiment and the first embodiment include that the image side surface **S2** of the first lens **L1** is convex near the optical axis, the image side surface **S8** of the fourth lens **L4** is convex near the optical axis, and the refractive power of the sixth lens **L6** is negative.

TABLE 2A

Surface numeral	Name	Surface type	Y radius (mm)	Thickness (mm)	Material	Refractive index	Abbe number	Focal length (mm)
	Object side	sphere	infinity	infinity				
S1	first lens L1	asphere	64.960	2.207	plastic	1.588	28.400	27.792
S2		asphere	-21.716	2.331				
S3	second lens L2	sphere	-11.248	1.180	glass	1.800	46.500	-11.872
S4		sphere	67.754	A				
S5(STO)	third lens L3	asphere	49.587	4.500	glass	1.678	55.600	9.746
S6	cemented surface	sphere	-7.370	0.005	cement			
S7	fourth lens L4	sphere	-7.370	1.036	glass	1.850	30.000	-14.622
S8		sphere	-19.110	1.761				
S9	fifth lens L5	asphere	15.301	4.917	plastic	1.544	56.000	15.620
S10		asphere	-17.077	B				
S11	sixth lens L6	asphere	78.040	1.795	plastic	1.544	56.000	-384.475
S12		asphere	56.426	1.480				
S13	seventh lens L7	asphere	-13.307	3.050	plastic	1.670	19.200	274.316
S14		asphere	-13.567	0.456				
S15	eighth lens L8	asphere	22.354	1.009	plastic	1.544	56.000	-18.107
S16		asphere	6.745	C				
S17	filter IR	sphere	infinity	0.300	glass	1.510	64.000	—
S18		sphere	infinity	1.662				
IMG	imaging plane	sphere	infinity	0.000				

TABLE 2B

	FNO	TTL(mm)	FOV(°)	ImgH(mm)	f(mm)	A(mm)	B(mm)	C(mm)
Short focal state (Z1)	2.200	40.800	38.300	5.720	16.200	5.641	5.125	2.277
Medium telephoto state (Z2)	2.600	40.800	31.800	5.720	19.400	3.900	4.300	4.845
Telephoto state (Z3)	3.000	40.800	29.600	5.720	20.800	3.138	4.125	5.778

[0122] Table 2C shows the high-order coefficients that applied for the aspherical surfaces in the second embodiment, and the surface shape of each lens is limited by the formula mentioned in the first embodiment.

TABLE 2C

Surface numeral	k	A4	A6	A8	A10
S1	4.435628E+01	1.320537E-04	-2.027949E-06	-4.013760E-07	1.065839E-07
S2	-5.780056E+00	1.160662E-04	-1.760192E-05	2.935022E-06	-2.901904E-07
S5	0.000000E+00	-2.607610E-05	-6.185391E-08	0.000000E+00	0.000000E+00
S9	-7.260294E+00	3.421711E-04	1.865967E-05	-2.680121E-06	2.825142E-07
S10	-1.769200E+01	-1.527615E-04	1.560160E-05	1.985347E-06	-3.553666E-07
S11	9.900000E+01	-2.102444E-04	8.430185E-05	-5.922854E-06	1.633925E-07
S12	-9.900000E+01	-6.561624E-04	1.631856E-04	-1.730313E-05	7.198756E-07
S13	3.967480E+00	-1.151617E-04	4.710965E-04	-1.172585E-04	1.547705E-05
S14	3.117892E+00	-1.303794E-03	1.186856E-03	-2.988502E-04	3.695965E-05
S15	-2.447444E+01	-1.091182E-02	3.175515E-03	-6.979949E-04	8.031371E-05
S16	-7.274847E+00	-6.942097E-03	1.862450E-03	-3.769170E-04	4.506010E-05

Surface numeral	A12	A14	A16	A18	A20
S1	-9.600855E-09	4.639012E-10	-1.247693E-11	1.753984E-13	-9.908977E-16
S2	1.889091E-08	-8.059689E-10	2.224347E-11	-3.632060E-13	2.677364E-15
S5	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S9	-1.862057E-08	7.940940E-10	-2.133124E-11	3.291729E-13	-2.212923E-15
S10	2.887429E-08	-1.269224E-09	2.974259E-11	-3.136038E-13	8.155969E-16
S11	-1.518777E-09	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S12	-1.039772E-08	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S13	-1.251337E-06	6.488943E-08	-2.092451E-09	3.801021E-11	-2.964268E-13
S14	-2.591624E-06	1.105784E-07	-2.950984E-09	4.798589E-11	-3.808327E-13
S15	-4.844947E-06	1.352528E-07	-2.459341E-10	-6.923089E-11	1.077327E-12
S16	-3.211790E-06	1.386194E-07	-3.549405E-09	4.956003E-11	-2.904632E-13

[0123] FIG. 2B shows the aberration diagram of the optical system 10 of the second embodiment in the short focal state Z1, FIG. 2D shows the aberration diagram of the optical system 10 of the second embodiment in the medium telephoto state Z2, and FIG. 2F shows the aberration diagram of the optical system 10 of the second embodiment in the telephoto state Z3. The detailed information can refer to the first embodiment. From the aberration diagrams in FIGS.

2B, 2D, and 2F, the longitudinal spherical aberration, the field curvature, and the distortion of the optical system 10 have been controlled, such that the optical system 10 of the embodiment has good imaging quality.

Third Embodiment

[0124] Referring to FIGS. 3A to 3D, the differences between the structures of the third embodiment and the first embodiment include that the refractive power of the sixth lens L6 is negative.

TABLE 3A

Surface numeral	Name	Surface type	Y radius (mm)	Thickness (mm)	Material	Refractive index	Abbe number	Focal length (mm)
S1	Object side	sphere	infinity	infinity				
S2	first lens L1	asphere	16.901	2.430	plastic	1.588	28.400	30.265
S3	second lens L2	asphere	288.511	1.949	glass	1.800	46.500	-12.480
S4		sphere	-15.927	0.883				
S5(STO)	third lens L3	asphere	28.082	A	glass	1.678	55.600	10.112
S6	cemented surface	sphere	13.341	3.595				
S7	fourth lens L4	sphere	-12.647	0.005	cement			
S8	fourth lens L4	sphere	-12.647	1.029	glass	1.850	30.000	-13.331
S9		sphere	120.113	0.531				
S10	fifth lens L5	asphere	13.386	4.890	plastic	1.544	56.000	14.885
S11	sixth lens L6	asphere	-18.010	B	plastic	1.544	56.000	-200.191
S12		asphere	92.090	1.772				
S13	seventh lens L7	asphere	49.638	1.450	plastic	1.670	19.200	72.464
S14		asphere	-13.972	3.675				
S15	eighth lens L8	asphere	-12.032	0.479	plastic	1.544	56.000	-16.185
S16		asphere	14.557	1.239				
S17	filter IR	sphere	5.333	C	glass	1.510	64.000	—
S18	imaging plane	sphere	infinity	0.300				
IMG		sphere	infinity	1.662				
		sphere	infinity	0.000				

TABLE 3B

	FNO	TTL(mm)	FOV(°)	ImgH(mm)	f(mm)	A(mm)	B(mm)	C(mm)
Short focal state (Z1)	2.000	39.200	34.600	5.720	18.500	6.236	5.182	1.895
Medium telephoto state (Z2)	2.500	39.200	26.000	5.720	24.600	3.544	4.153	5.614
Telephoto state (Z3)	2.900	39.200	21.000	5.720	30.700	1.168	4.232	7.910

[0125] Table 3C shows the high-order coefficients that applied for the aspherical surfaces in the third embodiment, and the surface shape of each lens is limited by the formula mentioned in the first embodiment.

TABLE 3C

Surface numeral	k	A4	A6	A8	A10
S1	1.134077E-01	8.464781E-05	3.813780E-07	3.358576E-07	-4.723290E-08
S2	-3.353937E-13	2.809722E-05	6.262983E-07	2.352728E-07	-4.107489E-08
S5	0.000000E+00	-3.272688E-05	-1.120010E-06	0.000000E+00	0.000000E+00
S9	-7.672276E+00	4.173731E-04	-8.325811E-07	-1.783809E-07	5.955542E-08
S10	-1.837692E+01	-5.401722E-05	2.138606E-05	-2.717400E-06	4.260179E-07
S11	9.900000E+01	3.107769E-04	-1.139815E-05	-4.107966E-07	3.835827E-08
S12	1.561939E+01	1.504025E-04	-1.610142E-05	-1.980262E-06	1.526624E-07
S13	4.065791E+00	6.742197E-04	3.061924E-06	-3.843481E-06	3.113485E-07
S14	2.508964E+00	-2.625736E-03	8.874169E-04	-1.494359E-04	1.688367E-05
S15	-6.536150E+01	-1.143863E-02	1.782846E-03	-2.470289E-04	2.495526E-05
S16	-8.178300E+00	-5.595390E-03	7.276023E-04	-8.632546E-05	7.709783E-06

Surface numeral	A12	A14	A16	A18	A20
S1	4.188567E-09	-2.143523E-10	6.435008E-12	-1.047118E-13	7.212886E-16
S2	4.944902E-09	-3.311482E-10	1.268311E-11	-2.582535E-13	2.192250E-15
S5	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S9	-5.601741E-09	3.043818E-10	-9.631677E-12	1.655002E-13	-1.190463E-15
S10	-4.144078E-08	2.508083E-09	-9.062343E-11	1.792110E-12	-1.483083E-14
S11	-7.513512E-10	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S12	-2.599414E-09	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S13	2.129260E-09	-1.113025E-09	5.690839E-11	-1.264846E-12	1.031423E-14
S14	-1.310356E-06	7.006304E-08	-2.466615E-09	5.116683E-11	-4.669659E-13
S15	-1.787399E-06	9.030783E-08	-3.103949E-09	6.488393E-11	-6.108621E-13
S16	-4.827466E-07	2.040736E-08	-5.552378E-10	8.774469E-12	-6.111900E-14

[0126] FIG. 3B shows the aberration diagram of the optical system 10 of the third embodiment in the short focal state Z1, FIG. 3D shows the aberration diagram of the optical system 10 of the third embodiment in the medium telephoto state Z2, and FIG. 3F shows the aberration diagram of the optical system 10 of the third embodiment in the telephoto state Z3. The detailed information can refer to the first embodiment. From the aberration diagrams in FIGS. 3B, 3D, and 3F, the longitudinal spherical aberration, the field curvature, and the distortion of the optical system 10 have been controlled, such that the optical system 10 of the embodiment has good imaging quality.

Fourth Embodiment

[0127] Referring to FIGS. 4A to 4F, the differences between the structures of the fourth embodiment and the first embodiment include that the image side surface S2 of the first lens L1 is convex near the optical axis, the image side surface S8 of the fourth lens L4 is convex near the optical axis, and the refractive power of the seventh lens L7 is negative.

TABLE 4A

Surface numeral	Name	Surface type	Y radius (mm)	Thickness (mm)	Material	Refractive index	Abbe number	Focal length (mm)
S1	Object side	sphere	infinity	infinity				
S2	first lens L1	asphere	55.754	2.190	plastic	1.588	28.400	28.326
S3		asphere	-23.587	2.102				
S4	second lens L2	sphere	-10.916	0.800	glass	1.800	46.500	-12.053
S5		sphere	92.403	A				
S5(STO)	third lens L3	asphere	44.110	4.479	glass	1.678	55.600	9.749
S6	cemented surface	sphere	-7.482	0.005	cement			
S7								
S7	fourth lens L4	sphere	-7.482	1.000	glass	1.850	30.000	-14.380
S8		sphere	-20.287	1.526				

TABLE 4A-continued

Surface numeral	Name	Surface type	Y radius (mm)	Thickness (mm)	Material	Refractive index	Abbe number	Focal length (mm)
S9	fifth lens L5	asphere	15.221	4.874	plastic	1.544	56.000	15.513
S10		asphere	-16.932	B				
S11	sixth lens L6	asphere	71.703	1.781	plastic	1.544	56.000	1548.63
S12		asphere	77.660	1.482				
S13	seventh lens L7	asphere	-12.405	3.242	plastic	1.670	19.200	-713.539
S14		asphere	-14.075	0.451				
S15	eighth lens L8	asphere	13.134	1.141	plastic	1.544	56.000	-18.401
S16		asphere	5.517	C				
S17	filter IR	sphere	infinity	0.300	glass	1.510	64.000	—
S18		sphere	infinity	1.662				
IMG	imaging plane	sphere	infinity	0.000				

TABLE 4B

	FNO	TTL(mm)	FOV(°)	ImgH(mm)	f(mm)	A(mm)	B(mm)	C(mm)
Short focal state (Z1)	2.200	40.000	39.000	5.720	16.100	5.698	5.071	2.183
Medium telephoto state (Z2)	2.500	40.000	33.200	5.720	18.900	4.242	4.256	4.452
Telephoto state (Z3)	2.900	40.000	32.000	5.720	19.600	3.885	4.127	4.941

[0128] Table 4C shows the high-order coefficients that applied for the aspherical surfaces in the fourth embodiment, and the surface shape of each lens is limited by the formula mentioned in the first embodiment.

TABLE 4C

Surface numeral	k	A4	A6	A8	A10
S1	5.027056E+01	1.344057E-04	-6.613175E-06	1.312633E-06	-1.265383E-07
S2	-6.376317E+00	1.184484E-04	-1.836870E-05	3.848205E-06	-4.506893E-07
S5	0.000000E+00	-5.056773E-05	6.134906E-08	0.000000E+00	0.000000E+00
S9	-7.150696E+00	3.677378E-04	1.033828E-05	-1.158172E-06	1.245798E-07
S10	-1.741326E+01	-1.656562E-04	2.550524E-05	-1.045014E-07	-1.391925E-07
S11	9.900000E+01	-1.905881E-04	1.134884E-04	-9.208279E-06	3.093171E-07
S12	-9.900000E+01	-8.217632E-04	1.993443E-04	-2.099915E-05	8.765692E-07
S13	3.938416E+00	6.864588E-04	1.007130E-04	-3.312322E-05	4.151573E-06
S14	3.505668E+00	-1.386340E-03	1.066074E-03	-2.613498E-04	3.241298E-05
S15	-3.702872E+01	-1.000551E-02	2.234966E-03	-4.094753E-04	3.601219E-05
S16	-9.200100E+00	-4.809888E-03	1.025715E-03	-1.955489E-04	2.219943E-05

Surface numeral	A12	A14	A16	A18	A20
S1	7.720447E-09	-2.985042E-10	7.225531E-12	-1.001346E-13	6.174905E-16
S2	3.273672E-08	-1.489492E-09	4.163082E-11	-6.548454E-13	4.467881E-15
S5	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S9	-8.411953E-09	3.584715E-10	-9.191846E-12	1.301208E-13	-7.788035E-16
S10	1.793207E-08	-1.150720E-09	4.221872E-11	-8.351960E-13	6.984510E-15
S11	-3.783293E-09	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S12	-1.279491E-08	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S13	-2.918449E-07	1.323579E-08	-3.813097E-10	6.306268E-12	-4.587591E-14
S14	-2.315731E-06	1.020780E-07	-2.846768E-09	4.857366E-11	-4.011629E-13
S15	-9.268775E-07	-7.508945E-08	6.558470E-09	-1.915806E-10	2.018205E-12
S16	-1.493207E-06	6.041671E-08	-1.442039E-09	1.868444E-11	-1.012702E-13

[0129] FIG. 4B shows the aberration diagram of the optical system 10 of the fourth embodiment in the short focal state Z1, FIG. 4D shows the aberration diagram of the optical system 10 of the fourth embodiment in the medium telephoto state Z2, and FIG. 4F shows the aberration diagram of the optical system 10 of the fourth embodiment in the telephoto state Z3. The detailed information can refer to the first embodiment. From the aberration diagrams in FIGS. 4B, 4D, and 4F, the longitudinal spherical aberration, the

field curvature, and the distortion of the optical system 10 have been controlled, such that the optical system 10 of the embodiment has good imaging quality.

Fifth Embodiment

[0130] Referring to FIGS. 5A to 5F, the differences between the structures of the fifth embodiment and the first embodiment include that the refractive power of the sixth lens L6 is negative.

TABLE 5A

Surface numeral	Name	Surface type	Y radius (mm)	Thickness (mm)	Material	Refractive index	Abbe number	Focal length (mm)
S1	Object side	sphere	infinity	infinity				
S2	first lens L1	asphere	17.220	2.421	plastic	1.588	28.400	31.680
S3		asphere	202.559	2.173				
S4	second lens L2	sphere	-15.014	0.800	glass	1.800	46.500	-12.582
S5		sphere	32.131	A				
S5(STO)	third lens L3	asphere	13.056	3.715	glass	1.678	55.600	9.947
S6	cemented surface	sphere	-12.422	0.005	cement			
S7	fourth lens L4	sphere	-12.422	1.000	glass	1.850	30.000	-12.907
S8		sphere	102.772	0.050				
S9	fifth lens L5	asphere	12.930	4.978	plastic	1.544	56.000	14.208
S10		asphere	-16.752	B				
S11	sixth lens L6	asphere	66.639	1.749	plastic	1.544	56.000	-234.882
S12		asphere	43.444	1.433				
S13	seventh lens L7	asphere	-12.972	3.045	plastic	1.670	19.200	54.535
S14		asphere	-10.512	0.511				
S15	eighth lens L8	asphere	9.632	1.040	plastic	1.544	56.000	-12.472
S16		asphere	3.837	C				
S17	filter IR	sphere	infinity	0.300	glass	1.510	64.000	—
S18		sphere	infinity	1.662				
IMG	imaging plane	sphere	infinity	0.000				

TABLE 5B

	FNO	TTL(mm)	FOV(°)	ImgH(mm)	f(mm)	A(mm)	B(mm)	C(mm)
Short focal state (Z1)	1.900	37.950	36.400	5.720	17.500	6.431	5.142	1.484
Medium telephoto state (Z2)	2.400	37.950	26.300	5.720	24.900	3.565	3.978	5.512
Telephoto state (Z3)	2.800	37.950	20.400	5.720	32.500	0.929	4.142	7.984

[0131] Table 5C shows the high-order coefficients that applied for the aspherical surfaces in the fourth embodiment,

and the surface shape of each lens is limited by the formula mentioned in the first embodiment.

TABLE 5C

Surface numeral	k	A4	A6	A8	A10
S1	1.313726E-01	1.377641E-04	-2.391413E-05	5.220392E-06	-5.770035E-07
S2	-9.850511E+01	-3.270553E-05	6.465166E-06	1.605572E-06	-3.905675E-07
S5	0.000000E+00	-3.835721E-05	-1.167966E-06	0.000000E+00	0.000000E+00
S9	-7.329075E+00	4.188334E-04	1.221470E-07	-1.202795E-07	2.924408E-08
S10	-1.711157E+01	-7.795059E-06	-1.137969E-05	5.515413E-06	-7.614913E-07
S11	9.898967E+01	3.018683E-04	-1.786809E-05	-4.732763E-08	5.859962E-08
S12	-4.046626E+01	2.553672E-04	-5.640294E-05	-4.358352E-07	2.057928E-07
S13	4.204622E+00	1.252781E-03	-1.195837E-04	7.813966E-06	-6.924012E-07
S14	2.485841E+00	-3.449871E-03	1.435454E-03	-3.146404E-04	4.490333E-05
S15	-7.051379E+01	-1.714881E-02	2.917575E-03	-4.682452E-04	5.768178E-05
S16	-1.320525E+01	-2.919362E-03	-1.976456E-04	7.699765E-05	-9.497048E-06
Surface numeral	A12	A14	A16	A18	A20
S1	3.823959E-08	-1.548639E-09	3.766481E-11	-5.049874E-13	2.875360E-15
S2	3.899609E-08	-2.123660E-09	6.616105E-11	-1.107592E-12	7.755779E-15
S5	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S9	-2.037823E-09	8.795972E-11	-2.234369E-12	3.078273E-14	-1.733113E-16
S10	6.388971E-08	-3.319020E-09	1.051792E-10	-1.861513E-12	1.418989E-14
S11	-1.472770E-09	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S12	-5.181663E-09	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
S13	8.658281E-08	-6.373051E-09	2.610391E-10	-5.637264E-12	4.983696E-14
S14	-4.208444E-06	2.579731E-07	-9.949579E-09	2.186217E-10	-2.075976E-12
S15	-4.936818E-06	2.816317E-07	-1.025514E-08	2.166147E-10	-2.019210E-12
S16	6.873293E-07	-3.150139E-08	8.943839E-10	-1.425372E-11	9.668977E-14

[0132] FIG. 5B shows the aberration diagram of the optical system 10 of the fifth embodiment in the short focal state Z1, FIG. 5D shows the aberration diagram of the optical system 10 of the fifth embodiment in the medium telephoto state Z2, and FIG. 5F shows the aberration diagram of the optical system 10 of the fifth embodiment in the telephoto state Z3. The detailed information can refer to the first embodiment. From the aberration diagrams in FIGS. 5B, 5D, and 5F, the longitudinal spherical aberration, the field curvature, and the distortion of the optical system 10 have been controlled, such that the optical system 10 of the embodiment has good imaging quality.

[0133] Table 6 shows the values of multiple relationships in the optical systems of the first to fifth embodiments.

[0134] The optical systems of the above embodiments have the characteristics of continuous zooming and high imaging quality.

[0135] Referring to FIG. 6, a camera module 20 is further provided according to an embodiment of the present disclosure. The camera module 20 includes the optical system 10 as mentioned in any above embodiment and a photosensitive chip 201. The photosensitive chip 201 is located on the image side of the optical system 10, and may be fixed to the optical system 10 by a bracket. The photosensitive chip 201 may be a complementary metal oxide semiconductor (CMOS) or a charge coupled device (CCD). Generally, the imaging surface IMG of the optical system 10 overlaps with the photosensitive surface of the photosensitive chip 201 after assembly. By incorporating the optical system 10 into

TABLE 6

Formula	First embodiment	Second embodiment	Third embodiment	Fourth embodiment	Fifth embodiment
f1/fmax	0.993	1.336	0.986	1.445	0.975
f2/fmax	-0.409	-0.571	-0.407	-0.615	-0.387
f3/fmax	0.337	0.469	0.329	0.497	0.306
f4/fmax	-0.441	-0.703	-0.434	-0.734	-0.397
f5/fmax	0.490	0.751	0.485	0.791	0.437
f6/fmax	37.619	-18.484	-6.521	79.012	-7.227
f7/fmax	3.054	13.188	2.360	-36.405	1.678
f8/fmax	-0.512	-0.871	-0.527	-0.939	-0.384
R1/fmax	0.576	3.123	0.551	2.845	0.530
R2/fmax	30.041	-1.044	9.398	-1.203	6.233
R3/fmax	-0.519	-0.541	-0.519	-0.557	-0.462
R4/fmax	0.926	3.257	0.915	4.714	0.989
R5/fmax	0.439	2.384	0.435	2.251	0.402
R6/fmax	-0.417	-0.354	-0.412	-0.382	0.382
R7/fmax	-0.417	-0.354	-0.412	-0.382	-0.382
R8/fmax	4.066	-0.919	3.912	-1.035	3.162
R9/fmax	0.440	0.736	0.436	0.777	0.398
R10/fmax	-0.595	-0.821	-0.587	-0.864	-0.515
R11/fmax	3.054	3.752	3.000	3.658	2.050
R12/fmax	3.563	2.713	1.617	3.962	1.337
R13/fmax	-0.395	-0.640	-0.455	-0.633	-0.399
R14/fmax	-0.374	-0.652	-0.392	-0.718	-0.323
R15/fmax	0.616	1.075	0.474	0.670	0.296
R16/fmax	0.187	0.324	0.174	0.281	0.118
f12/fmax	-0.87	-1.21	-0.88	-1.27	-0.81
f345/fmax	0.39	0.54	0.38	0.57	0.34
f678/fmax	-0.56	-0.82	-0.56	-0.85	-0.43
fmax/fmin	1.631	1.284	1.659	1.217	1.857
f12/f345	-2.255	-2.230	-2.329	-2.214	-2.363
f12/f678	1.561	1.480	1.575	1.502	1.887
FOVmax/FOVmin	1.633	1.294	1.648	1.219	1.784
FOVmax	34.300	38.300	34.600	39.000	36.400
FOVmin	21.000	29.600	21.000	32.000	20.400
FNOmax/FNOmin	1.364	1.364	1.450	1.318	1.474
FNOmax	3.000	3.000	2.900	2.900	2.800
FNOmin	2.200	2.200	2.000	2.200	1.900
TTL/ImgH	7.010	7.133	6.853	6.993	6.635
Yc82/SD82	0.627	0.742	0.652	0.680	0.632
SDmax/ImgH	1.014	1.014	1.080	1.036	1.103
GL2/GL1	2.114	2.137	1.910	2.334	1.807
GL2/GL3	1.249	1.569	1.167	1.468	1.254
TDmax/TDmin	1.189	1.106	1.205	1.083	1.232
ΔA	4.941	2.503	5.068	1.812	5.502
ΔC	5.739	3.501	6.015	2.758	6.500
ΔA/ΔC	0.8609	0.7149	0.8425	0.6571	0.8465
CT1/CT2	2.745	1.8708	2.7536	2.7371	3.0246
CT3/CT4	4.3488	4.3452	3.4924	4.4781	3.7155
CT5/CT4	4.8745	4.7477	4.7514	4.8728	4.9783
CT7/CT6	2.1491	1.6996	2.0741	1.8204	1.7413
CT7/CT8	2.8958	3.023	2.965	2.8422	2.9279

the camera module 20, the camera module 20 has the characteristics of continuous zooming and high imaging quality.

[0136] Referring to FIG. 7, an electronic device 30 is further provided according to an embodiment of the present disclosure. The electronic device 30 includes a housing 310 and the camera module 20 as mentioned above. The camera module 20 is mounted to the housing 310. The housing 310 may be a display screen, a circuit board, a middle frame, a back cover, and any other component. The electronic device 30 may be a smartphone, a smartwatch, a smart glass, an e-book reader, a tablet computer, a biometric device, or a PDA. Due to the continuous zooming and high imaging quality characteristics of the camera module 20, the camera module 20 may be assembled into a smaller space inside the electronic device 30, thereby reducing the thickness of the electronic device 30 and increasing the zooming speed.

[0137] The above embodiments are only for describing but not intended to limit the present disclosure. Although the embodiments of the present disclosure have been described, those having ordinary skill in the art can understand that changes may be made within the principles of the present disclosure, up to and including the full extent established by the broad general meaning of the terms used in the claims. It will, therefore, be appreciated that the embodiments described above may be modified within the scope of the claims.

What is claimed is:

1. An optical system consisting of eight lenses having refractive power, from an object side to an image side along an optical axis of the optical system, the eight lenses sequentially comprising a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens, wherein,

the first lens and the second lens form a first lens group with negative refractive power;

the third lens to the fifth lens form a second lens group with positive refractive power;

the sixth lens to the eighth lens form a third lens group having negative refractive power;

the first lens group is fixed relative to an imaging plane of the optical system, and each of the second lens group and the third lens group is configured to move along the optical axis between the first lens group and the imaging plane to realize zooming capability of the optical system;

the optical system satisfies following relationships:

$$1 < f_{\max} / f_{\min} < 2.1;$$

$$-1.5 < f_{12} / f_{\max} < -0.5;$$

$$0.25 < f_{345} / f_{\max} < 0.8; \text{ and}$$

$$-1.1 < f_{678} / f_{\max} < -0.3,$$

wherein, $f_{\max}$ is a maximum focal length of the optical system, $f_{\min}$ is a minimum focal length of the optical system, f_{12} is a combined focal length of the first lens to the second lens, f_{345} is a combined focal length of the third lens to the fifth lens, and f_{678} is a combined focal length of the sixth lens to the eighth lens.

2. The optical system according to claim 1, further satisfying at least one of following relationships:

$$1 < FOV_{\max} / FOV_{\min} < 2.1;$$

$$30 < FOV_{\max} < 45; \text{ and}$$

$$16 < FOV_{\min} < 36,$$

wherein, $FOV_{\max}$ is a maximum field of view of the optical system, and $FOV_{\min}$ is a minimum field of view of the optical system.

3. The optical system according to claim 1, further satisfying at least one of following relationships:

$$1 < FNO_{\max} / FNO_{\min} < 1.8;$$

$$2.4 < FNO_{\max} < 3.4; \text{ and}$$

$$1.5 < FNO_{\min} < 2.6,$$

wherein, $FNO_{\max}$ is a maximum F-number of the optical system, and $FNO_{\min}$ is a minimum F-number of the optical system.

4. The optical system according to claim 1, further satisfying at least one of following relationships:

$$-1.9 < f_{12} / f_{345} < -2.7; \text{ and}$$

$$1.2 < f_{12} / f_{678} < 2.1.$$

5. The optical system according to claim 1, further satisfying following relationship:

$$5 < TTL / ImgH < 9,$$

wherein, TTL is a distance from an object side surface of the first lens to the imaging plane at the optical axis, and ImgH is half of an image height corresponding to a maximum field of view of the optical system.

6. The optical system according to claim 1, further satisfying following relationship:

$$0.3 < Y_{c82} / SD_{82} < 0.9,$$

wherein, Y_{c82} is a vertical height from an off-axis vertex of an image side surface of the eighth lens to the optical axis, and SD_{82} is an effective half-aperture of the image side surface of the eighth lens.

7. The optical system according to claim 1, further satisfying following relationship:

$$0.8 < SD_{\max} / ImgH < 1.3,$$

wherein, $SD_{\max}$ is a maximum effective half-aperture from an object side surface of the first lens to an image side surface of the eighth lens, and ImgH is half of an image height corresponding to a maximum field of view of the optical system.

8. The optical system according to claim 1, further satisfying at least one of following relationships:

$$1.5 < GL2/GL1 < 2.7; \text{ and} \\ 0.9 < GL2/GL3 < 1.8,$$

wherein, GL1 is a distance between the object side surface of the first lens and the image side surface of the second lens at the optical axis, GL2 is a distance between an object side surface of the third lens and an image side surface of the fifth lens at the optical axis, and GL3 is a distance between an object side surface of the sixth lens and the image side surface of the eighth lens at the optical axis.

9. The optical system according to claim 1, further satisfying following relationship:

$$1 < TD_{\max}/TD_{\min} < 1.5,$$

wherein, TD_{max} is a maximum distance between the object side surface of the first lens and the image side surface of the eighth lens at the optical axis, and TD_{min} is minimum distance between the object side surface of the first lens and the image side surface of the eighth lens at the optical axis.

10. The optical system according to claim 1, further satisfying following relationship:

$$0.5 < \Delta A/\Delta C < 1.1,$$

wherein, ΔA is a maximum stroke of the second lens group at the optical axis, and ΔC is a maximum stroke of the third lens group at the optical axis.

11. The optical system according to claim 1, wherein the first lens has positive refractive power, and an object side surface of the first lens is convex near the optical axis;

the second lens has negative refractive power, an object side surface of the second lens is concave near the optical axis, and an image side surface of the second lens is concave near the optical axis;

the third lens has positive refractive power, an object side surface of the third lens is convex near the optical axis, and an image side surface of the third lens is convex near the optical axis;

the fourth lens has negative refractive power, and an object side surface of the fourth lens is concave near the optical axis;

the fifth lens has positive refractive power, an object side surface of the fifth lens is convex near the optical axis, and an image side surface of the fifth lens is convex near the optical axis;

the sixth lens has refractive power, an object side surface of the sixth lens is convex near the optical axis, and an image side surface of the sixth lens is concave near the optical axis;

the seventh lens has refractive power, an object side surface of the seventh lens is concave near the optical axis, and an image side surface of the seventh lens is convex near the optical axis; and

the eighth lens has negative refractive power, an object side surface of the eighth lens is convex near the optical axis, and an image side surface of the eighth lens is concave near the optical axis.

12. The optical system according to claim 1, further satisfying at least one of following relationships:

$$0.7 < f1/f_{\max} < 1.7; \\ -0.8 < f2/f_{\max} < -0.2; \\ 0.2 < f3/f_{\max} < 0.7; \\ -0.8 < f4/f_{\max} < -0.2; \\ 0.25 < f5/f_{\max} < 1; \text{ and} \\ 1.2 < f8/f_{\max} < -0.25,$$

wherein, f1 is a focal length of the first lens, f2 is a focal length of the second lens, f3 is a focal length of the third lens, f4 is a focal length of the fourth lens, f5 is a focal length of the fifth lens, and f8 is a focal length of the eighth lens.

13. The optical system according to claim 1, further satisfying at least one of following relationships:

$$0.4 < R1/f_{\max}; \\ 0.7 < R4/f_{\max}; \\ 0.2 < R5/f_{\max}; \\ -0.6 < R6/f_{\max} < -0.2; \\ 0.3 < R9/f_{\max} < 1; \\ -1.1 < R10/f_{\max} < -0.3; \\ 1.5 < R11/f_{\max}; \\ 1 < R12/f_{\max}; \\ -0.9 < R13/f_{\max} < -0.3; \\ -0.9 < R14/f_{\max} < -0.2; \\ 0.2 < R15/f_{\max} < 1.4; \text{ and} \\ 0 < R16/f_{\max} < 0.4,$$

wherein, R1 is a radius of curvature of an object side surface of the first lens at the optical axis, R4 is a radius of curvature of an image side surface of the second lens at the optical axis, R5 is a radius of curvature of an object side surface of the third lens at the optical axis, R6 is a radius of curvature of an image side surface of the third lens at the optical axis, R9 is a radius of curvature of an object side surface of the fifth lens at the optical axis, R10 is a radius of curvature of an image side surface of the fifth lens at the optical axis, R11 is a radius of curvature of an object side surface of the sixth lens at the optical axis, R12 is a radius of curvature of an object side surface of the sixth lens at the optical axis, R13 is a radius of curvature of an object side surface of the seventh lens at the optical axis, R14 is a radius of curvature of an image side surface of the seventh lens at the optical axis, R15 is a radius of curvature of an object side surface of the eighth lens at the optical axis, and R16 is a radius of curvature of an image side surface of the eighth lens at the optical axis.

14. The optical system according to claim **1**, further satisfying at least one of following relationships:

$$\begin{aligned} 1.4 < CT1/CT2 < 3.5; \\ 3 < CT3/CT4 < 5; \\ 4 < CT5/CT4 < 5.5; \\ 1.1 < CT7/CT6 < 2.8; \text{ and} \\ 2.2 < CT7/CT8 < 3.7, \end{aligned}$$

wherein, CT1 is a thickness of the first lens at the optical axis, CT2 is a thickness of the second lens at the optical axis, CT3 is a thickness of the third lens at the optical axis, CT4 is a thickness of the fourth lens at the optical axis, CT5 is a thickness of the fifth lens at the optical axis, CT6 is a thickness of the sixth lens at the optical axis, CT7 is a thickness of the seventh lens at the optical axis, and CT8 is a thickness of the eighth lens at the optical axis.

15. A camera module comprising:

an optical system according to claim **1**; and
a photosensitive chip located on the image side of the optical system.

16. The camera module according to claim **15**, wherein the optical system further satisfies at least one of following relationships:

$$1 < FOV_{\max}/FOV_{\min} < 2.1;$$

$$30 < FOV_{\max} < 45; \text{ and}$$

$$16 < FOV_{\min} < 36,$$

wherein, FOV_{max} is a maximum field of view of the optical system, and FOV_{min} is a minimum field of view of the optical system.

17. The camera module according to claim **15**, wherein the optical system further satisfies at least one of following relationships:

$$1 < FNO_{\max}/FNO_{\min} < 1.8;$$

$$2.4 < FNO_{\max} < 3.4; \text{ and}$$

$$1.5 < FNO_{\min} < 2.6,$$

wherein, FNO_{max} is a maximum F-number of the optical system, and FNO_{min} is a minimum F-number of the optical system.

18. The camera module according to claim **15**, wherein the optical system further satisfies at least one of following relationships:

$$-1.9 < f_{12}/f_{345} < -2.7; \text{ and}$$

$$1.2 < f_{12}/f_{678} < 2.1.$$

19. The camera module according to claim **15**, wherein the optical system further satisfies at least one of following relationships:

$$5 < TTL/ImgH < 9;$$

$$0.3 < Yc82/SD82 < 0.9;$$

$$0.8 < SD_{\max}/ImgH < 1.3;$$

$$1.5 < GL2/GL1 < 2.7;$$

$$0.9 < GL2/GL3 < 1.8;$$

$$1 < TD_{\max}/TD_{\min} < 1.5; \text{ and}$$

$$0.5 < \Delta A/\Delta C < 1.1,$$

wherein, TTL is a distance from an object side surface of the first lens to the imaging plane at the optical axis, ImgH is half of an image height corresponding to a maximum field of view of the optical system, Yc82 is a vertical height from an off-axis vertex of an image side surface of the eighth lens to the optical axis, SD82 is an effective half-aperture of the image side surface of the eighth lens, SD_{max} is a maximum effective half-aperture from the object side surface of the first lens to the image side surface of the eighth lens, GL1 is a distance between the object side surface of the first lens and the image side surface of the second lens at the optical axis, GL2 is a distance between an object side surface of the third lens and an image side surface of the fifth lens at the optical axis, GL3 is a distance between an object side surface of the sixth lens and the image side surface of the eighth lens at the optical axis, TD_{max} is a maximum distance between the object side surface of the first lens and the image side surface of the eighth lens at the optical axis, TD_{min} is minimum distance between the object side surface of the first lens and the image side surface of the eighth lens at the optical axis, ΔA is a maximum stroke of the second lens group at the optical axis, ΔC is a maximum stroke of the third lens group at the optical axis.

20. An electronic device comprising:

a housing; and

a camera module according to claim **15**, wherein the camera module is received in the housing.

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